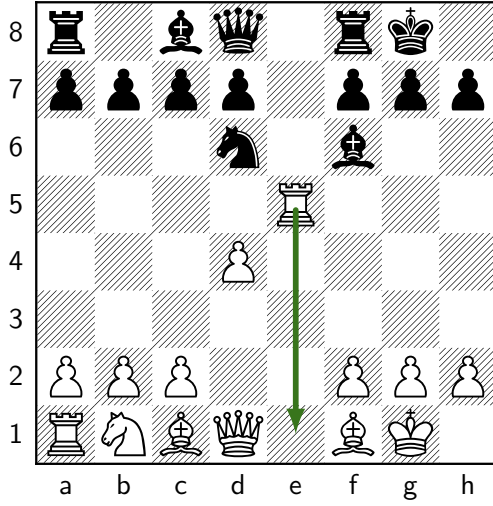


Melanie-BOTS (corpus: CCRL engines) — match game 4

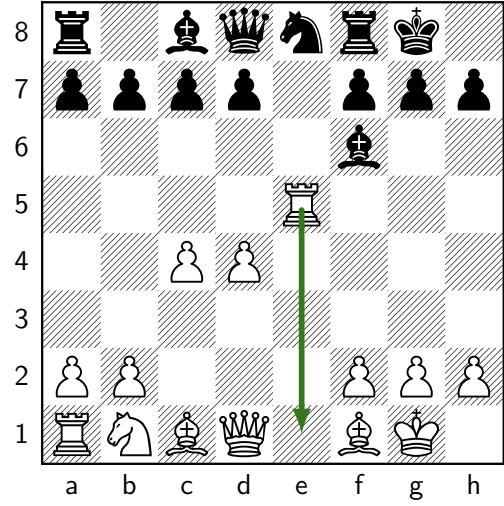
Melanie (Deep_Vision)

Abstract

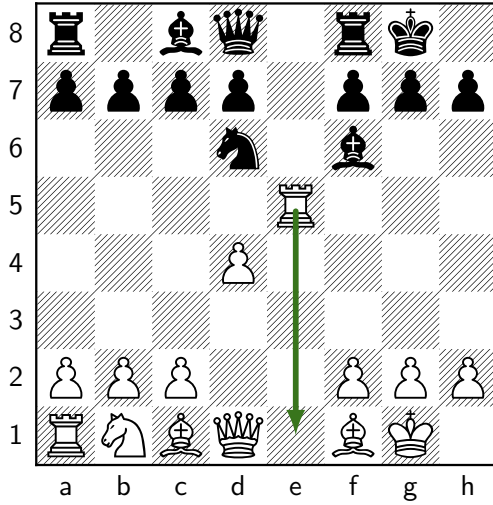
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



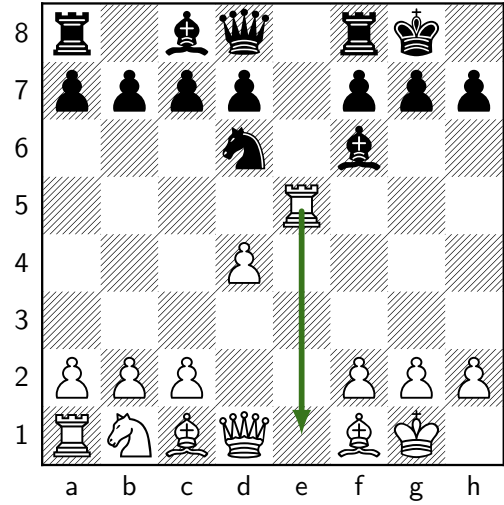
(a)



(b)

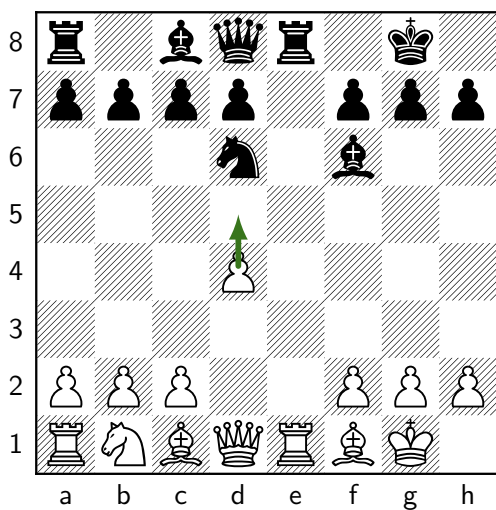


(c)

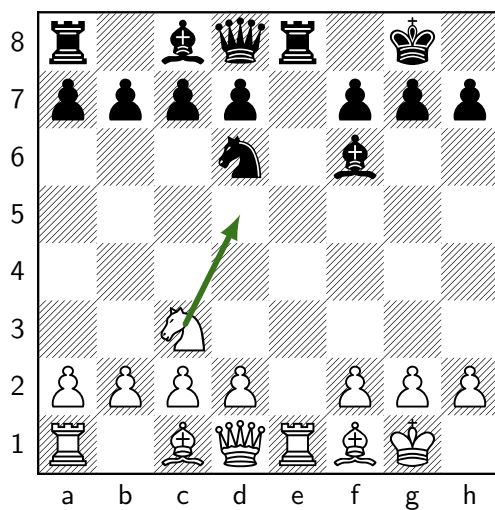


(d)

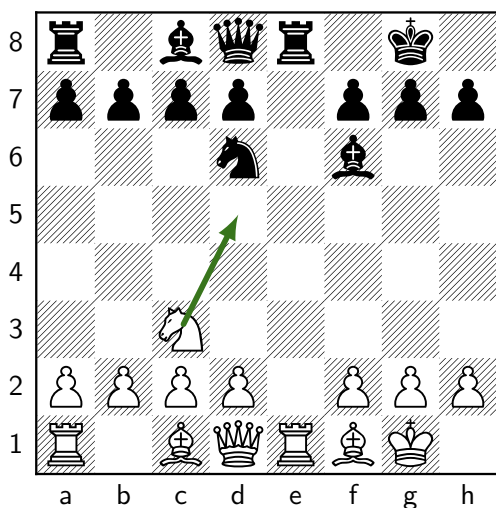
Figure 1: **Move 10, White: Re1.** (a) Our position; Melanie plays **Re1** (**e5e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#1311` (ply 20) [structural distance $d=21$, QAP obj=27]: the master played **Re1**; under μ our *Re5* plays the role of that game’s *Re5*; cross-type roles: our *Pg2* as their *Ra1*, our *Ph2* as their *Bf1*; transferred move **e5e1** (**endorses this move**). (c) Retrieved from `cctl_top_p0001#1075` (ply 18) [structural distance $d=0$, QAP obj=26]: the master played **Re1**; under μ our *Re5* plays the role of that game’s *Re5*; cross-type roles: our *nd6* as their *Ra1*, our *Pg2* as their *Qd1*; transferred move **e5e1** (**endorses this move**). (d) Retrieved from `cctl_top_p0002#784` (ply 18) [structural distance $d=0$, QAP obj=26]: the master played **Re1**; under μ our *Re5* plays the role of that game’s *Re5*; cross-type roles: our *nd6* as their *Ra1*, our *Pg2* as their *Qd1*; transferred move **e5e1** (**endorses this move**). Synthesis: 16 retrieved analogs transferred to **Re1** (summed QAP objective 407, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



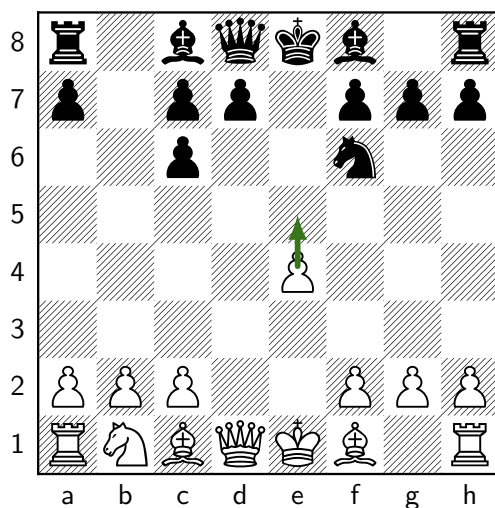
(a)



(b)

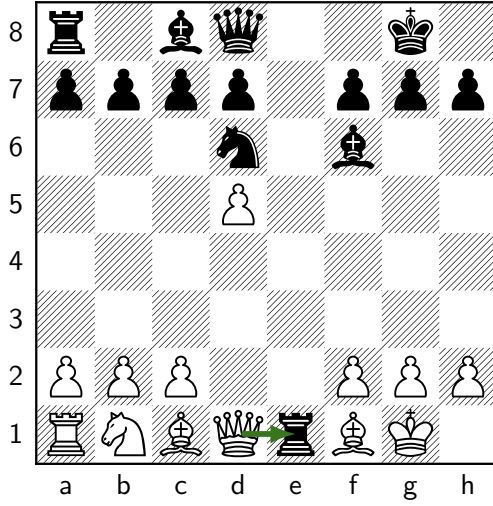


(c)

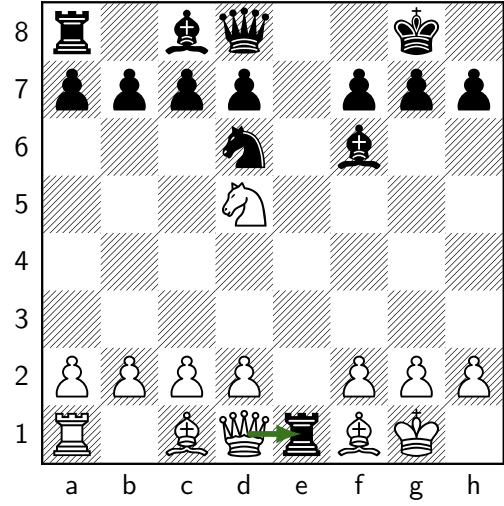


(d)

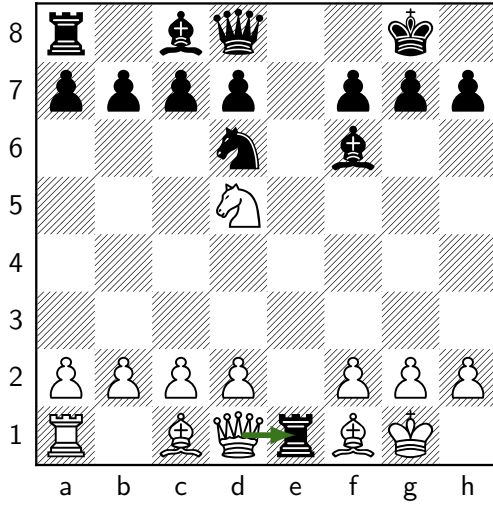
Figure 2: **Move 11, White: d5.** (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1677` (ply 20) [structural distance $d=38$, QAP obj=55]: the master played Nd5; under μ our $Pd4$ plays the role of that game’s $Nc3$ (a **cross-type** correspondence); cross-type roles: our $Bc1$ as their $Ra1$, our $Pg2$ as their $Bc1$; transferred move d4d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3182` (ply 20) [structural distance $d=38$, QAP obj=55]: the master played Nd5; under μ our $Pd4$ plays the role of that game’s $Nc3$ (a **cross-type** correspondence); cross-type roles: our $Bc1$ as their $Ra1$, our $Pg2$ as their $Bc1$; transferred move d4d5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3128` (ply 10) [structural distance $d=69$, QAP obj=21]: the master played e5; under μ our $Pd4$ plays the role of that game’s $Pe4$; cross-type roles: our $Nb1$ as their $Ra1$, our $Bf1$ as their $Nb1$; transferred move d4d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to d5 (summed QAP objective 152, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



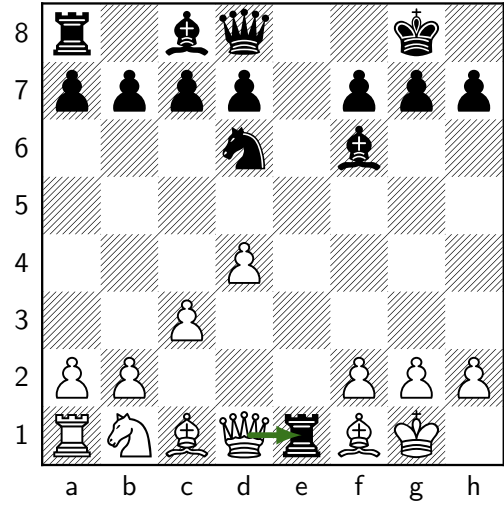
(a)



(b)

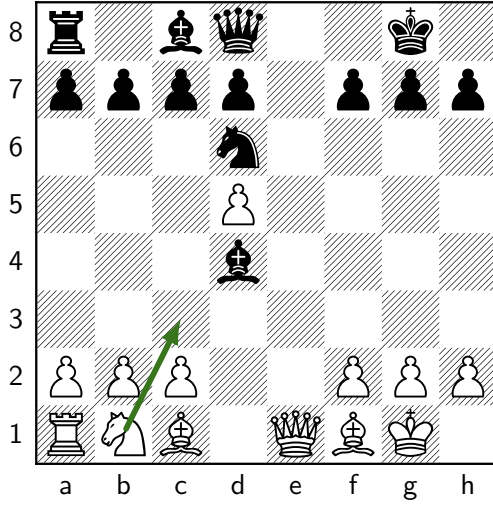


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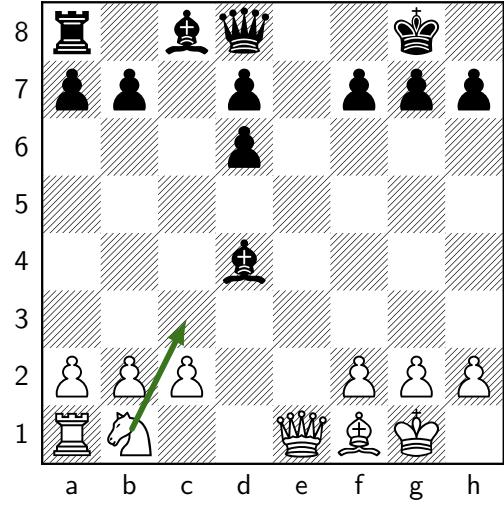


(d)

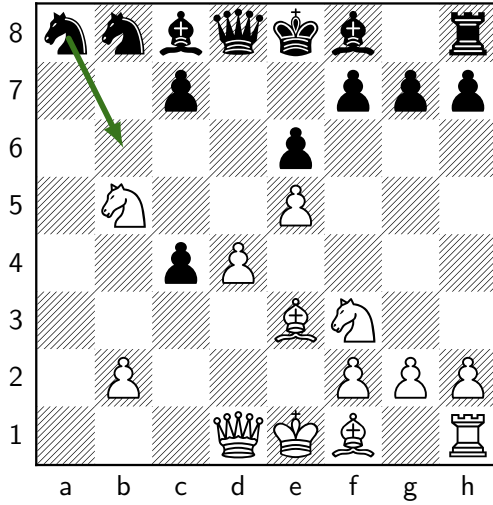
Figure 3: **Move 12, White: Qxe1.** (a) Our position; Melanie plays Qxe1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1677` (ply 22) [structural distance $d=32$, QAP obj=117]: the master played Qxe1; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our Nb1 as their Pa2, our bc8 as their nd6; transferred move d1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3182` (ply 22) [structural distance $d=32$, QAP obj=117]: the master played Qxe1; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our Nb1 as their Pa2, our bc8 as their nd6; transferred move d1e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#784` (ply 22) [structural distance $d=21$, QAP obj=109]: the master played Qxe1; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our Nb1 as their Ra1, our pg7 as their Nb1; transferred move d1e1 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Qxe1 (summed QAP objective 1207, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



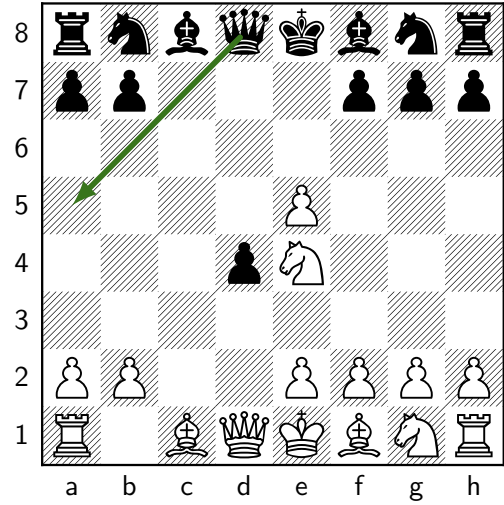
(a)



(b)

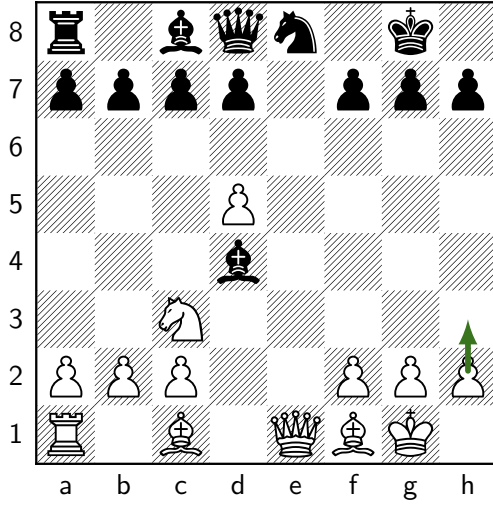


(c)

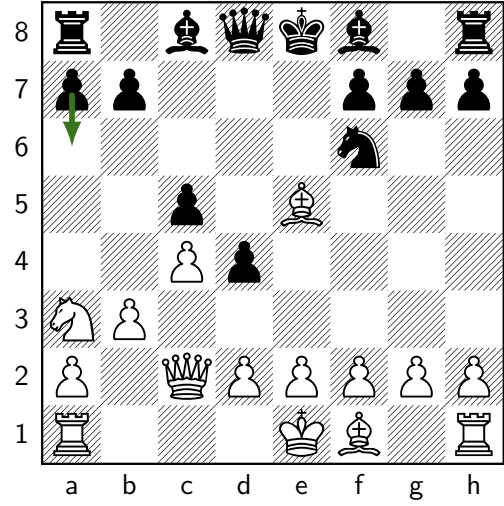


(d)

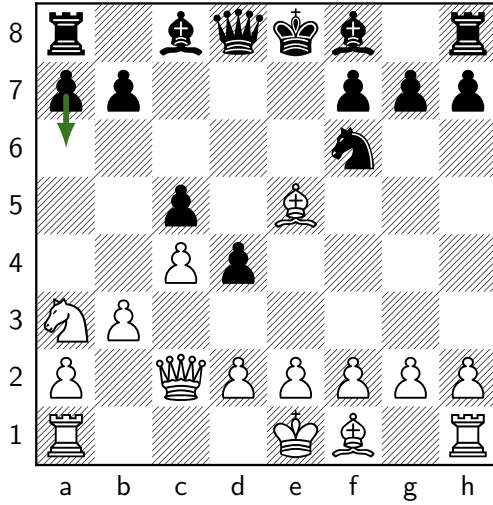
Figure 4: **Move 13, White: Nc3.** (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#702` (ply 26) [structural distance $d=81$, QAP obj=125]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Bc1 as their Pa2, our nd6 as their pd6; transferred move b1c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#339` (ply 21) [structural distance $d=72$, QAP obj=38]: the master played Nb6; under μ our Ph2 plays the role of that game’s Na1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Nb8, our pa7 as their Bc8; transferred move h2h3. (d) Retrieved from `ccr1_top_p0000#1945` (ply 11) [structural distance $d=83$, QAP obj=29]: the master played Qa5+; under μ our Bc1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Nb8, our pd7 as their Bc8; transferred move c1g5. Synthesis: 1 retrieved analog transferred to Nc3 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



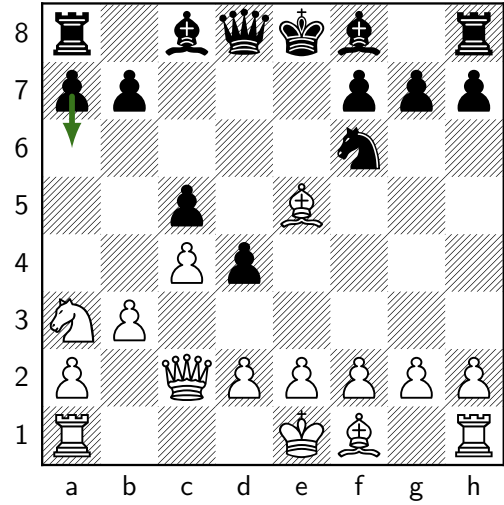
(a)



(b)

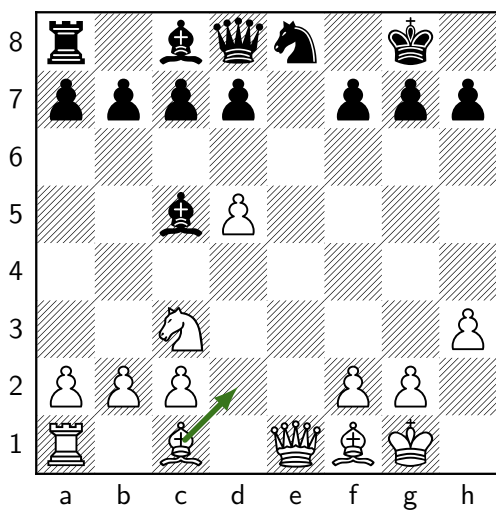


(c)

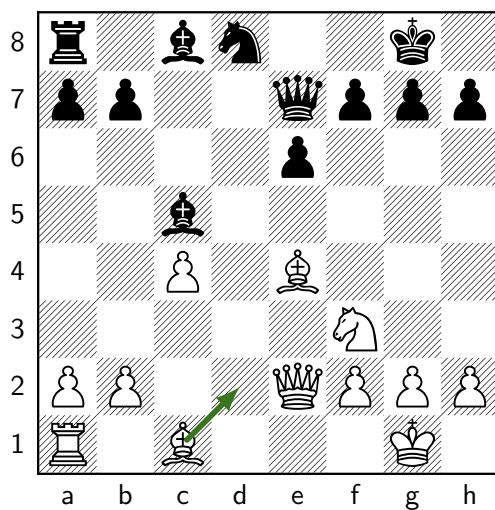


(d)

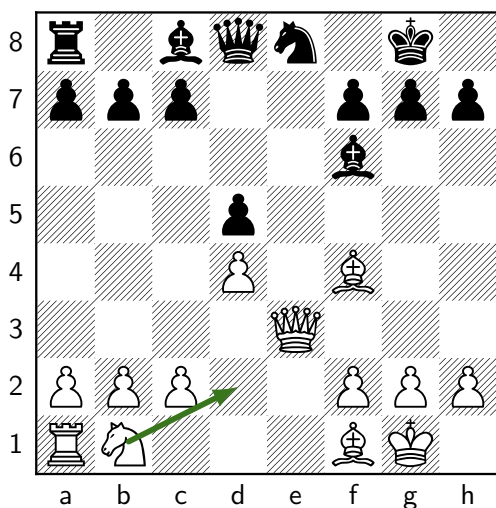
Figure 5: **Move 14, White: h3**. (a) Our position; Melanie plays **h3** (**h2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1907` (ply 15) [structural distance $d=95$, QAP obj=23]: the master played **a6**; under μ our $Ph2$ plays the role of that game’s $Pa2$; cross-type roles: our $Pd5$ as their $Ra8$, our $Pb2$ as their $Bf8$; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2660` (ply 15) [structural distance $d=95$, QAP obj=23]: the master played **a6**; under μ our $Ph2$ plays the role of that game’s $Pa2$; cross-type roles: our $Pd5$ as their $Ra8$, our $Pb2$ as their $Bf8$; transferred move **h2h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3189` (ply 15) [structural distance $d=95$, QAP obj=23]: the master played **a6**; under μ our $Ph2$ plays the role of that game’s $Pa2$; cross-type roles: our $Pd5$ as their $Ra8$, our $Pb2$ as their $Bf8$; transferred move **h2h3** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **h3** (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



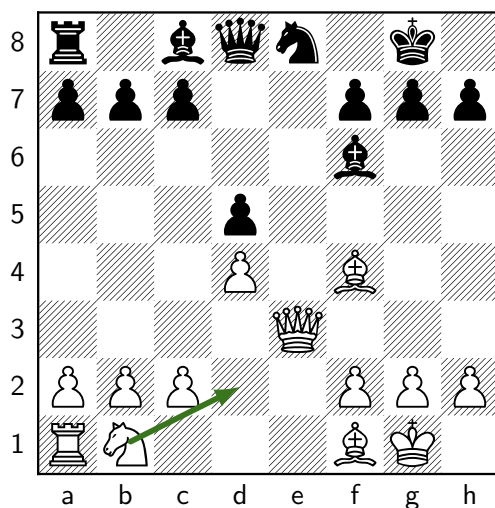
(a)



(b)

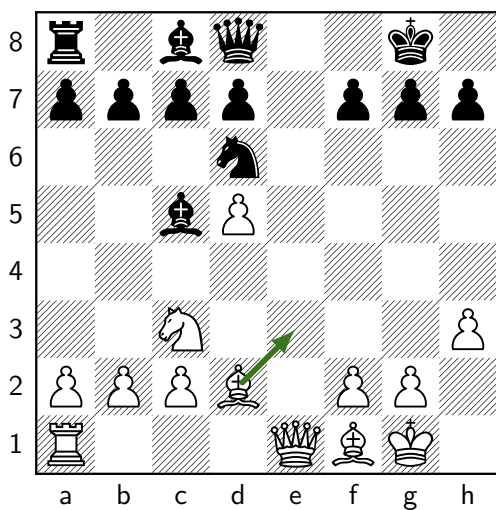


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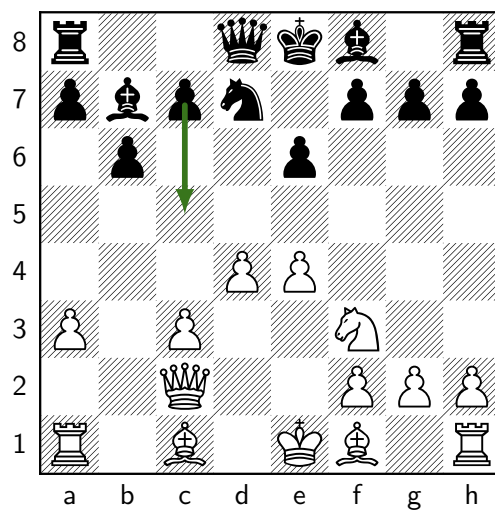


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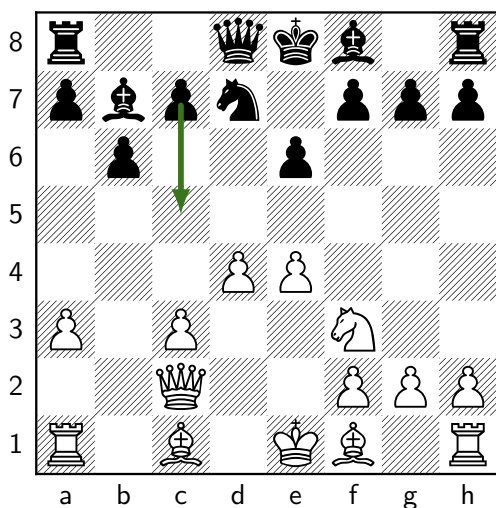
Figure 6: **Move 15, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3525` (ply 34) [structural distance $d=72$, QAP obj=80]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *pb7* as their *Ra1*, our *Ra1* as their *Pb2*; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3734` (ply 26) [structural distance $d=77$, QAP obj=29]: the master played Nd2; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pd5* as their *Ra1*, our *qd8* as their *Pb2*; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1312` (ply 26) [structural distance $d=77$, QAP obj=29]: the master played Nd2; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pd5* as their *Ra1*, our *qd8* as their *Pb2*; transferred move c1d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd2 (summed QAP objective 150, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



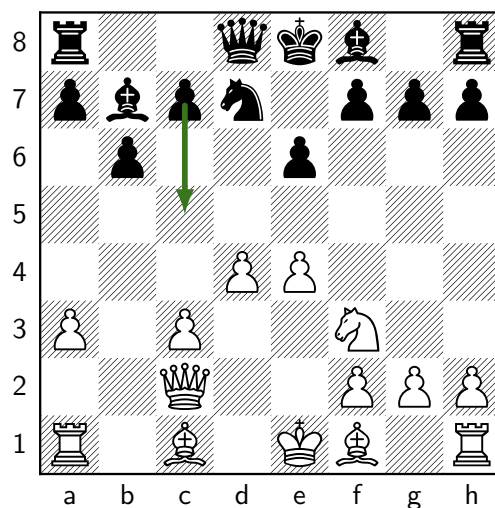
(a)



(b)

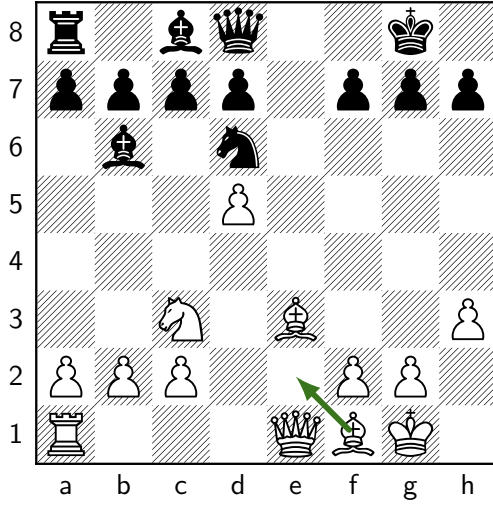


(c)

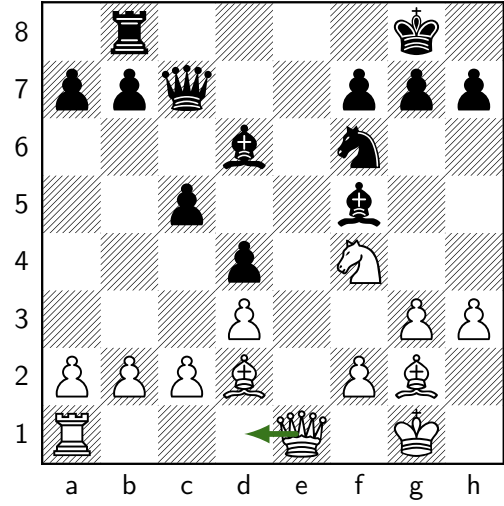


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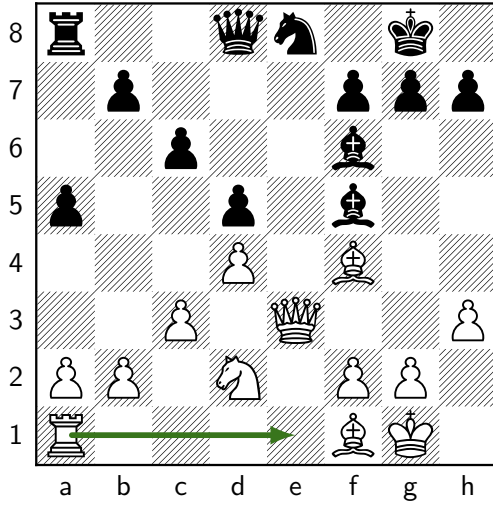
Figure 7: **Move 16, White: Be3**. (a) Our position; Melanie plays Be3 (d2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1969` (ply 17) [structural distance $d=75$, QAP obj=43]: the master played c5; under μ our *Bd2* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *nd6* as their *Ra8*, our *pc7* as their *Bf8*; transferred move d2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1970` (ply 17) [structural distance $d=75$, QAP obj=43]: the master played c5; under μ our *Bd2* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *nd6* as their *Ra8*, our *pc7* as their *Bf8*; transferred move d2e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1971` (ply 17) [structural distance $d=75$, QAP obj=43]: the master played c5; under μ our *Bd2* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *nd6* as their *Ra8*, our *pc7* as their *Bf8*; transferred move d2e3 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Be3 (summed QAP objective 768, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



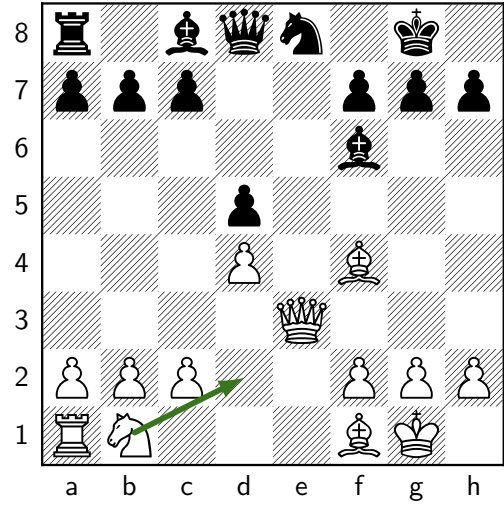
(a)



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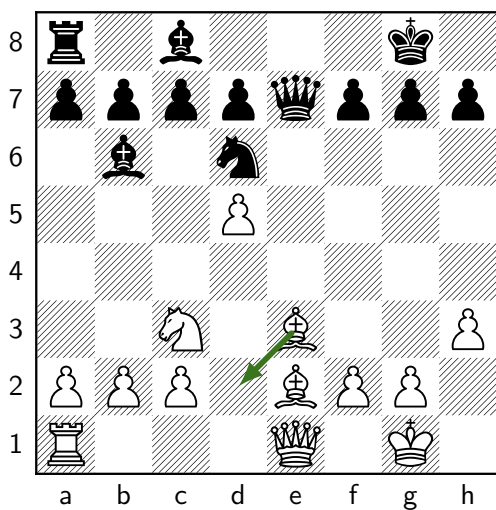


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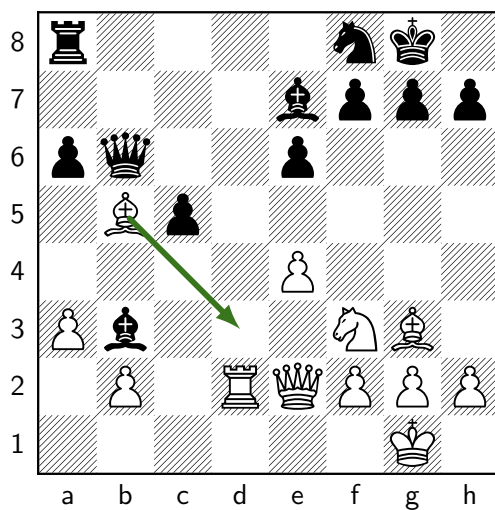


(d)

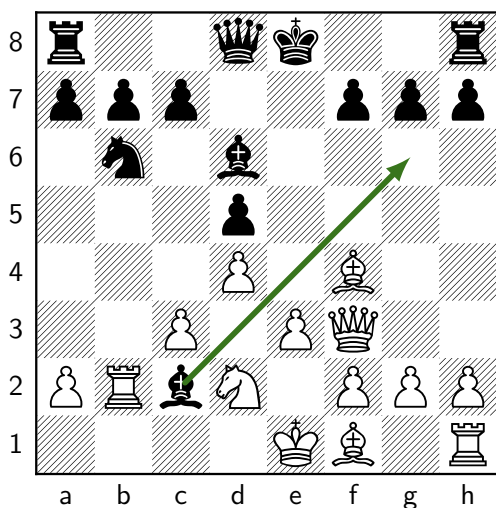
Figure 8: **Move 17, White: Be2**. (a) Our position; Melanie plays Be2 (f1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1083` (ply 32) [structural distance $d=92$, QAP obj=18]: the master played Qd1; under μ our Bf1 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Qe1 as their Pa2, our Nc3 as their Pb2; transferred move f1e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#741` (ply 32) [structural distance $d=75$, QAP obj=18]: the master played Re1; under μ our Bf1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Bf1, our bc8 as their Nd2; transferred move f1e2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3734` (ply 26) [structural distance $d=85$, QAP obj=17]: the master played Nd2; under μ our Bf1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our pd7 as their Bf1, our Be3 as their Qe3; transferred move f1e2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Be2 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



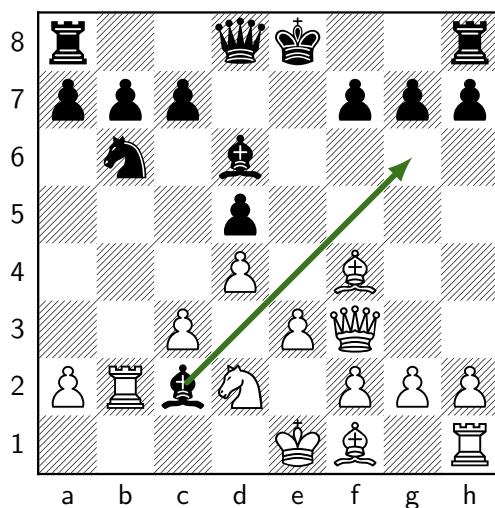
(a)



(b)

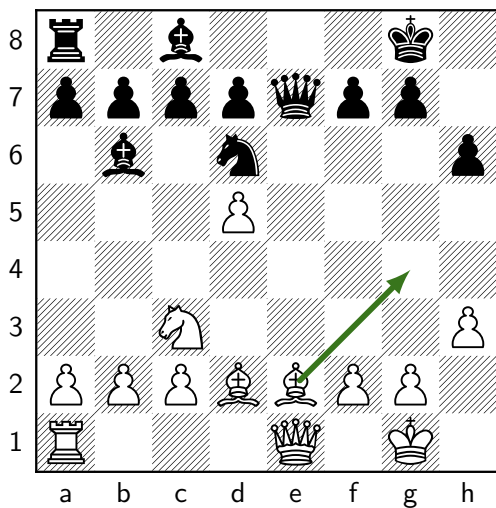


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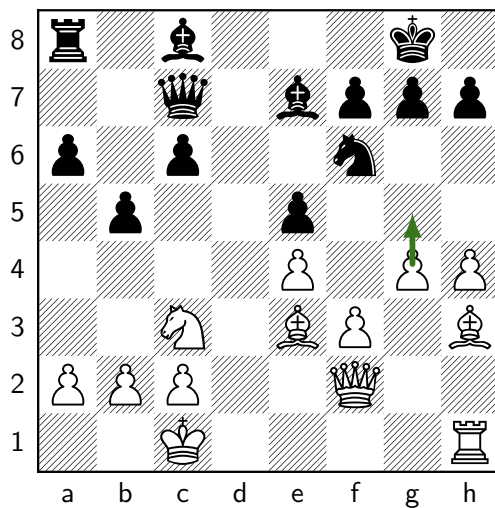


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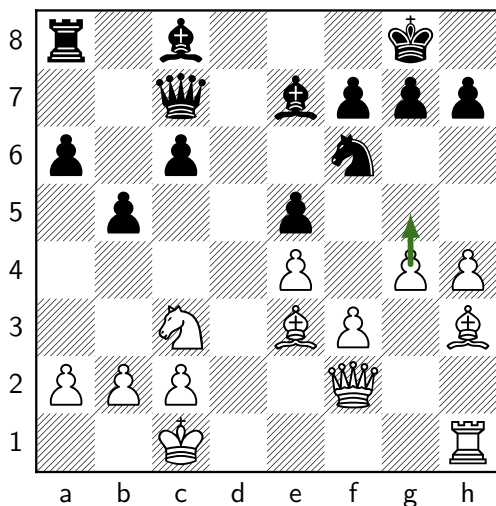
Figure 9: **Move 18, White: Bd2.** (a) Our position; Melanie plays Bd2 (e3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#79` (ply 40) [structural distance $d=88$, QAP obj=32]: the master played Bd3; under μ our *Be3* plays the role of that game’s *Bb5*; cross-type roles: our *Ra1* as their *Pb2*, our *Be2* as their *Rd2*; transferred move e3d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3095` (ply 27) [structural distance $d=90$, QAP obj=20]: the master played Bg6; under μ our *Be3* plays the role of that game’s *Bc7*; cross-type roles: our *Pf2* as their *Ra8*, our *Be2* as their *Rh8*; transferred move e3d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3096` (ply 27) [structural distance $d=90$, QAP obj=20]: the master played Bg6; under μ our *Be3* plays the role of that game’s *Bc7*; cross-type roles: our *Pf2* as their *Ra8*, our *Be2* as their *Rh8*; transferred move e3d2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bd2 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



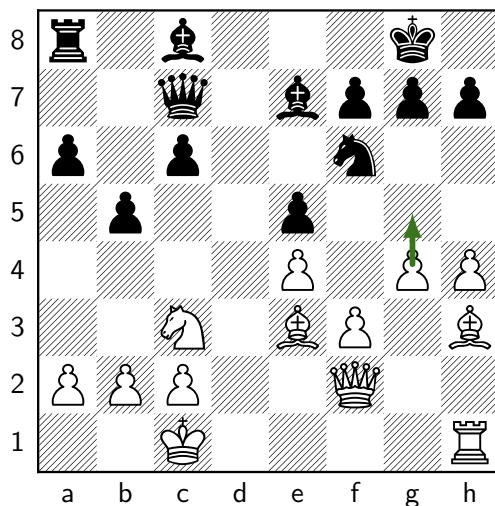
(a)



(b)

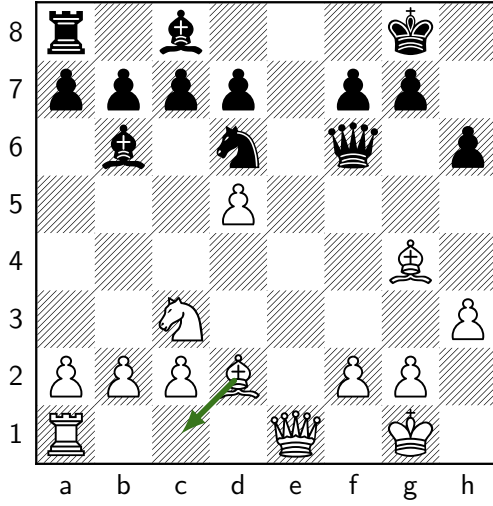


(c)

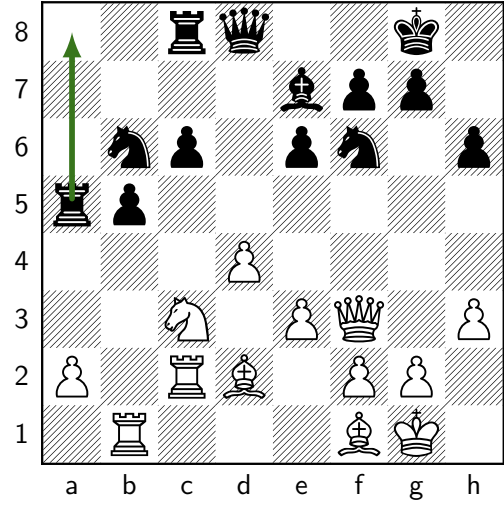


(d)

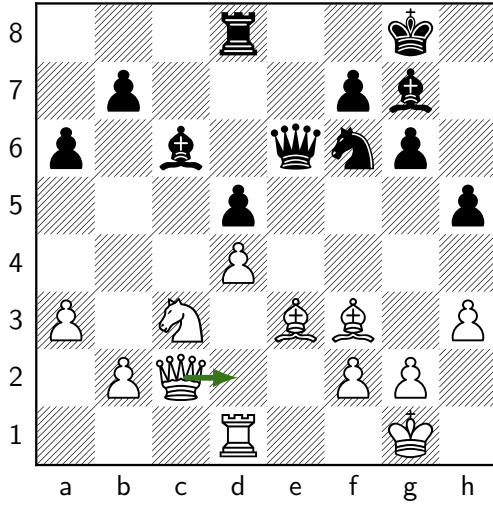
Figure 10: **Move 19, White: Bg4**. (a) Our position; Melanie plays Bg4 (e2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3422` (ply 30) [structural distance $d=96$, QAP obj=52]: the master played g5; under μ our *Be2* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Rh1*, our *Ra1* as their *Qf2*; transferred move e2g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3423` (ply 30) [structural distance $d=96$, QAP obj=52]: the master played g5; under μ our *Be2* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Rh1*, our *Ra1* as their *Qf2*; transferred move e2g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3480` (ply 30) [structural distance $d=96$, QAP obj=52]: the master played g5; under μ our *Be2* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Rh1*, our *Ra1* as their *Qf2*; transferred move e2g4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bg4 (summed QAP objective 245, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



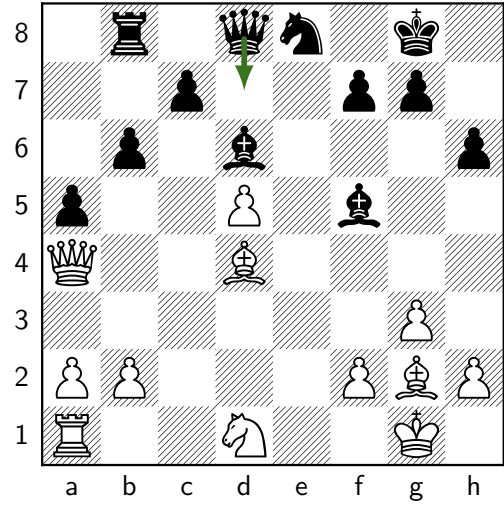
(a)



(b)

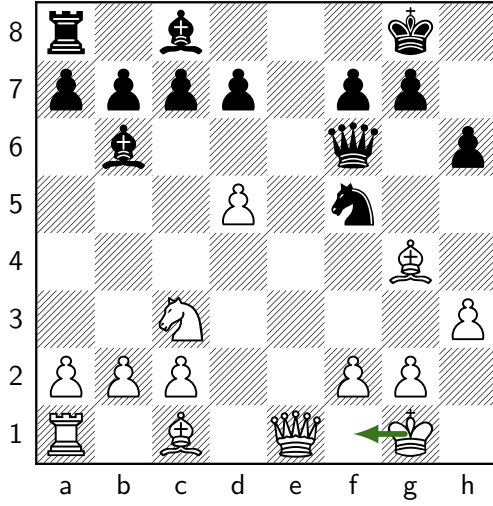


(c)

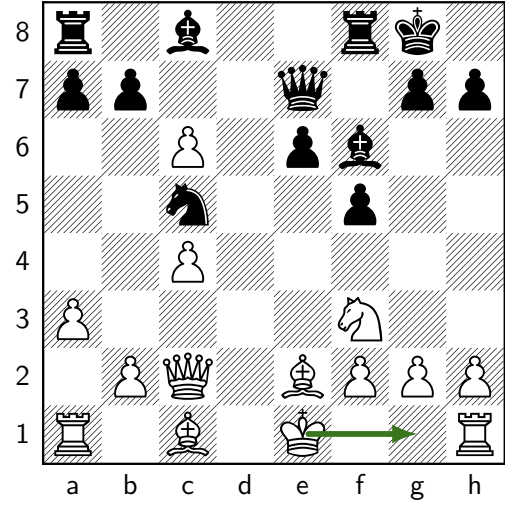


(d)

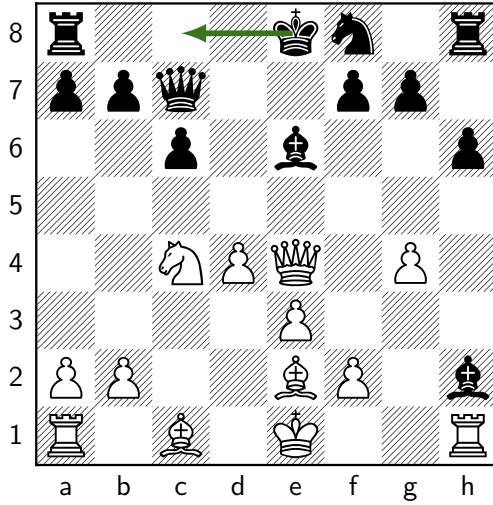
Figure 11: **Move 20, White: Bc1.** (a) Our position; Melanie plays Bc1 (d2c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#876` (ply 43) [structural distance $d=105$, QAP obj=74]: the master played Raa8; under μ our Bd2 plays the role of that game’s Ra4 (a **cross-type** correspondence); cross-type roles: our Qe1 as their Rc8, our Bg4 as their Qd8; transferred move d2c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1421` (ply 46) [structural distance $d=100$, QAP obj=71]: the master played Qd2; under μ our Bd2 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qe1 as their Rd1, our Ra1 as their Pb2; transferred move d2c1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#751` (ply 39) [structural distance $d=91$, QAP obj=46]: the master played Qd7; under μ our Bd2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Rb8, our Qe1 as their Ph6; transferred move d2c1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bc1 (summed QAP objective 327, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



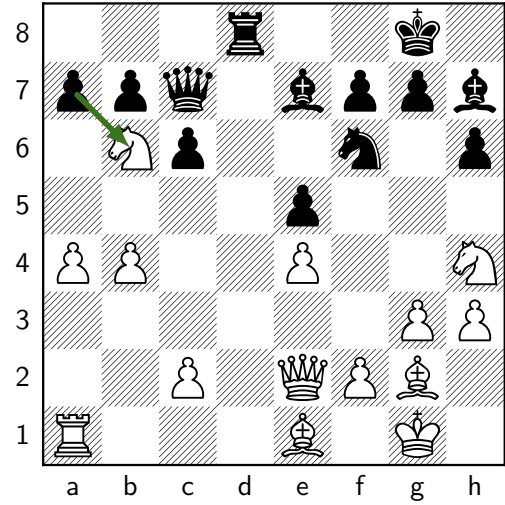
(a)



(b)

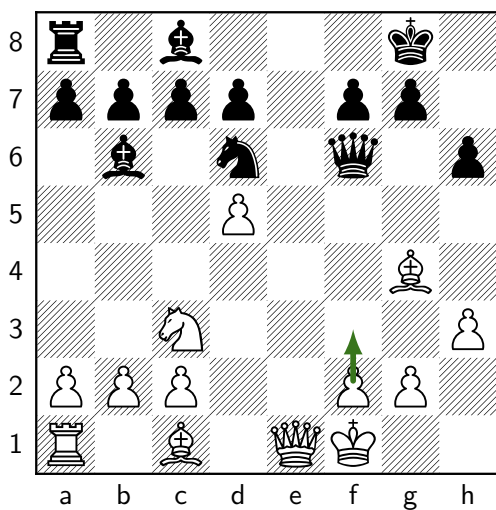


(c)

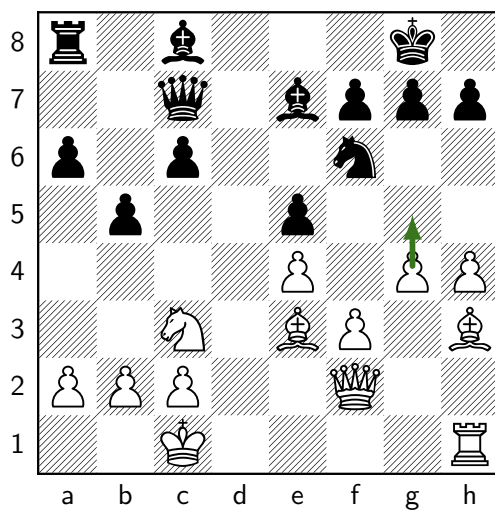


(d)

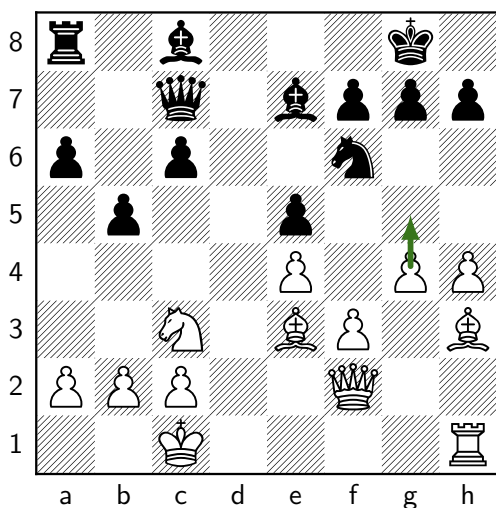
Figure 12: **Move 21, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1436` (ply 28) [structural distance $d=97$, QAP obj=31]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *nf5* as their *Ra1*, our *Qe1* as their *Bc1*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2510` (ply 31) [structural distance $d=81$, QAP obj=29]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qe1* as their *Ra8*, our *Pc2* as their *Nf8*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#607` (ply 37) [structural distance $d=87$, QAP obj=37]: the master played axb6; under μ our *Bg4* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Qe1* as their *Rd8*, our *Bg4* as their *Qc7*; transferred move g4e2. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



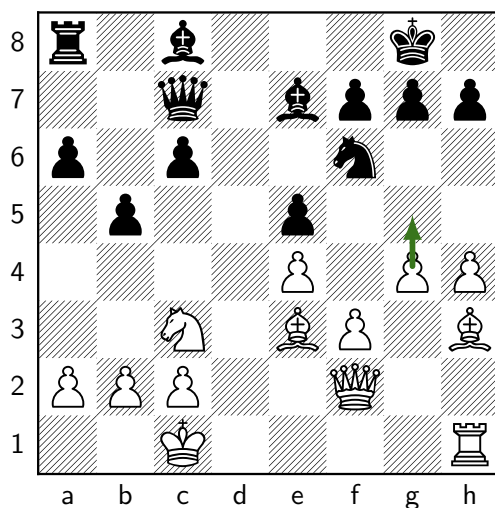
(a)



(b)

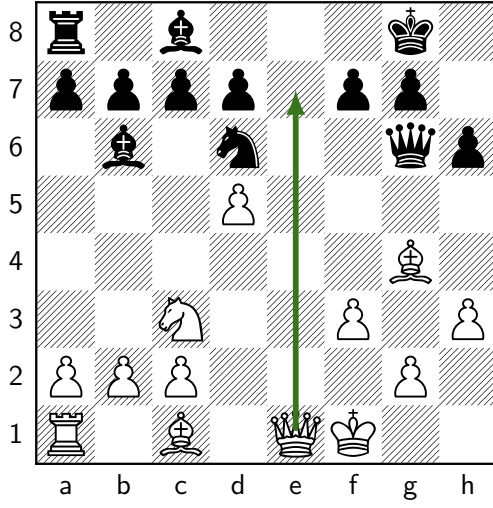


(c)

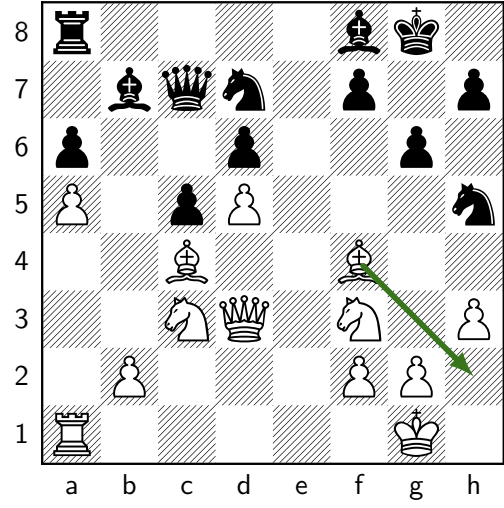


(d)

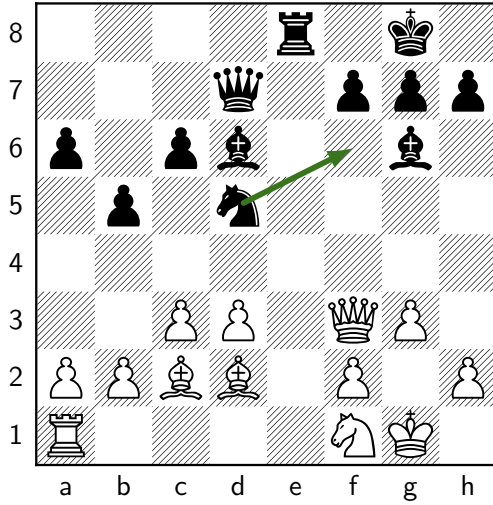
Figure 13: **Move 22, White: f3.** (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3422` (ply 30) [structural distance $d=108$, QAP obj=44]: the master played g5; under μ our $Pf2$ plays the role of that game’s $Pg4$; cross-type roles: our $Bc1$ as their $Rh1$, our $bb6$ as their $Pb2$; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3423` (ply 30) [structural distance $d=108$, QAP obj=44]: the master played g5; under μ our $Pf2$ plays the role of that game’s $Pg4$; cross-type roles: our $Bc1$ as their $Rh1$, our $bb6$ as their $Pb2$; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3480` (ply 30) [structural distance $d=108$, QAP obj=44]: the master played g5; under μ our $Pf2$ plays the role of that game’s $Pg4$; cross-type roles: our $Bc1$ as their $Rh1$, our $bb6$ as their $Pb2$; transferred move f2f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f3 (summed QAP objective 174, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



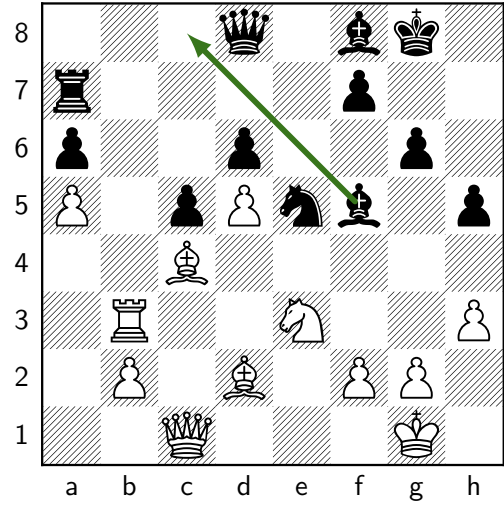
(a)



(b)

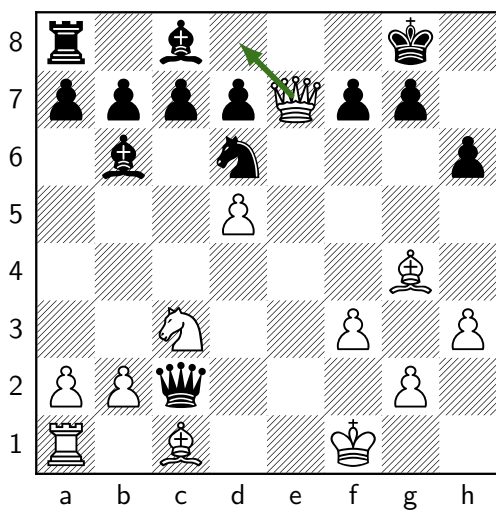


(c)

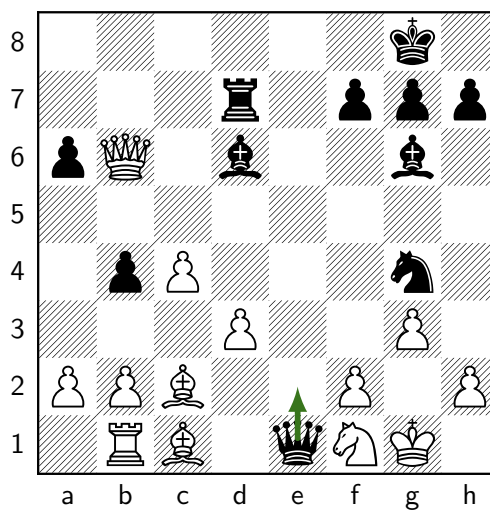


(d)

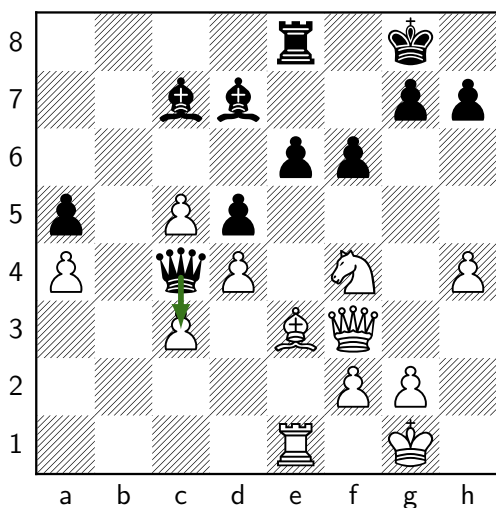
Figure 14: **Move 23, White: Qe7.** (a) Our position; Melanie plays Qe7 (e1e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1659` (ply 40) [structural distance $d=106$, QAP obj=77]: the master played Bh2; under μ our Bg4 plays the role of that game’s Bf4; cross-type roles: our ph6 as their Qd3, our Qe1 as their Nf3; transferred move g4h5. (c) Retrieved from `ccrl_top_p0002#1509` (ply 41) [structural distance $d=86$, QAP obj=68]: the master played Nf6; under μ our Bg4 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Qe1 as their Re8, our bb6 as their Qd7; transferred move g4f5. (d) Retrieved from `ccrl_top_p0000#838` (ply 53) [structural distance $d=113$, QAP obj=67]: the master played Bc8; under μ our Bg4 plays the role of that game’s Bf4; cross-type roles: our nd6 as their pa5, our qg6 as their Ne5; transferred move g4h5. Synthesis: no retrieved analog transferred to Qe7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



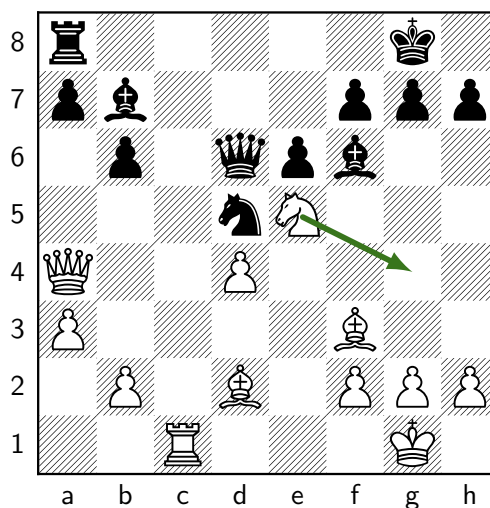
(a)



(b)

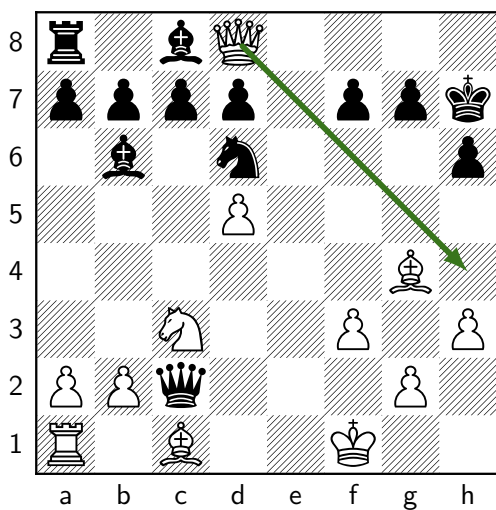


(c)

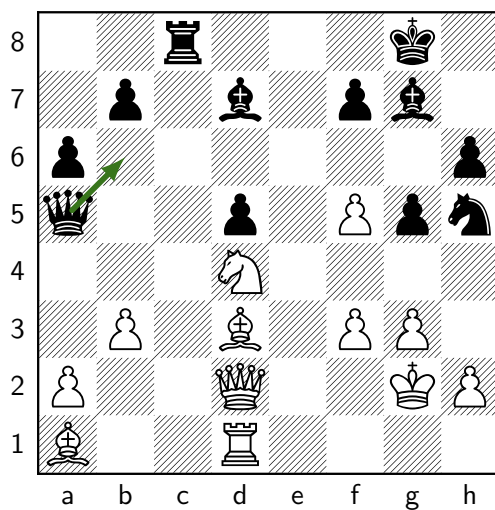


(d)

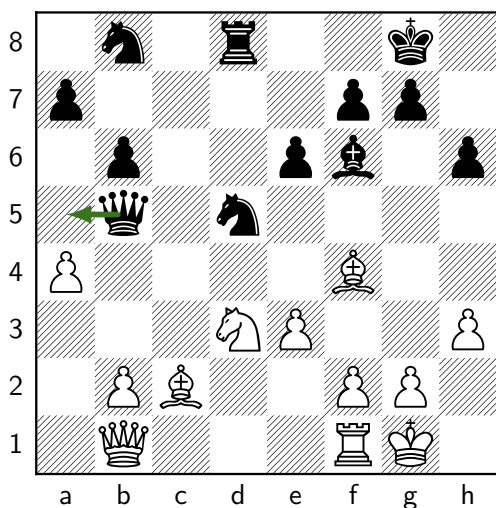
Figure 15: **Move 24, White: Qd8+.** (a) Our position; Melanie plays Qd8+ (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2630` (ply 45) [structural distance $d=103$, QAP obj=96]: the master played Qe2; under μ our Qe7 plays the role of that game’s Qe8; cross-type roles: our Pd5 as their Bg6, our Nc3 as their Pb4; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3405` (ply 57) [structural distance $d=134$, QAP obj=48]: the master played Qxc3; under μ our Qe7 plays the role of that game’s Qc5; cross-type roles: our Pf3 as their Re8, our qc2 as their Bc7; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3870` (ply 44) [structural distance $d=127$, QAP obj=69]: the master played Ng4; under μ our Bc1 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our Nc3 as their Rc1, our Ra1 as their Ph2; transferred move c1g5. Synthesis: 2 retrieved analogs transferred to Qd8+ (summed QAP objective 144, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



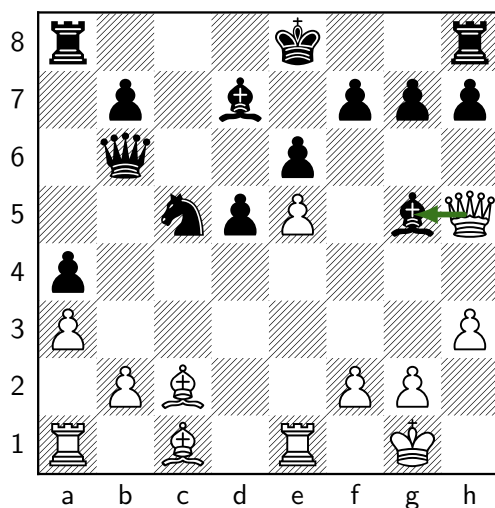
(a)



(b)

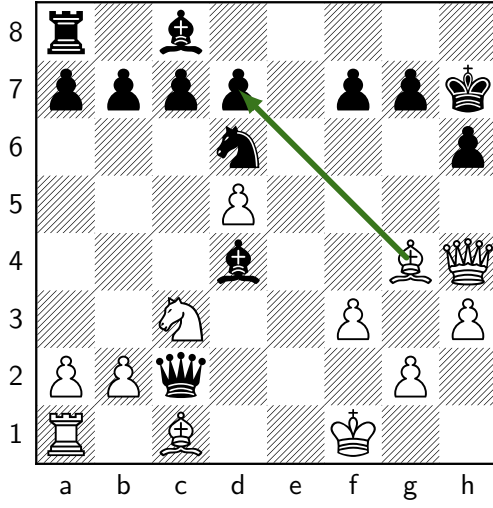


(c)

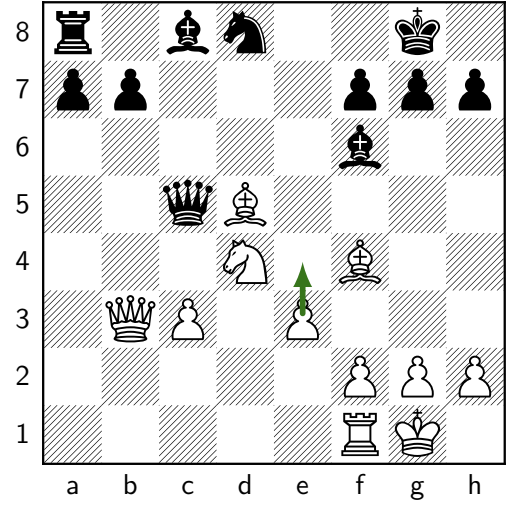


(d)

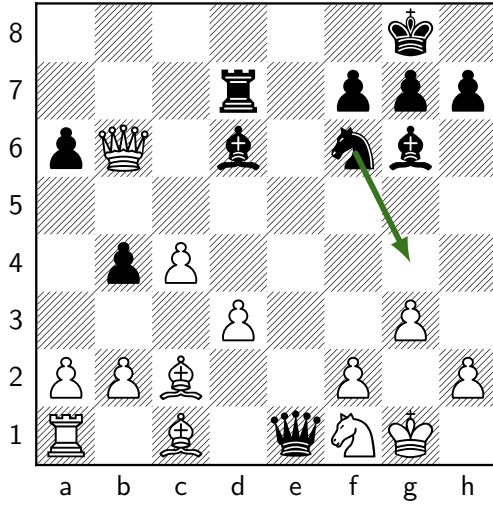
Figure 16: **Move 25, White: Qh4.** (a) Our position; Melanie plays **Qh4** (**d8h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2895` (ply 65) [structural distance $d=143$, QAP obj=44]: the master played **Qb6**; under μ our *Qd8* plays the role of that game’s *Qa4*; cross-type roles: our *Bg4* as their *Rc8*, our *pb7* as their *Bd7*; transferred move **d8h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#549` (ply 51) [structural distance $d=148$, QAP obj=41]: the master played **Qa5**; under μ our *Qd8* plays the role of that game’s *Qb4*; cross-type roles: our *Ra1* as their *Nb8*, our *Bc1* as their *Rd8*; transferred move **d8h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3114` (ply 34) [structural distance $d=142$, QAP obj=39]: the master played **Qxg5**; under μ our *Qd8* plays the role of that game’s *Qh5*; cross-type roles: our *Pd5* as their *Ra1*, our *Bg4* as their *Re1*; transferred move **d8h4** (**endorses this move**). Synthesis: 8 retrieved analogs transferred to **Qh4** (summed QAP objective 268, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



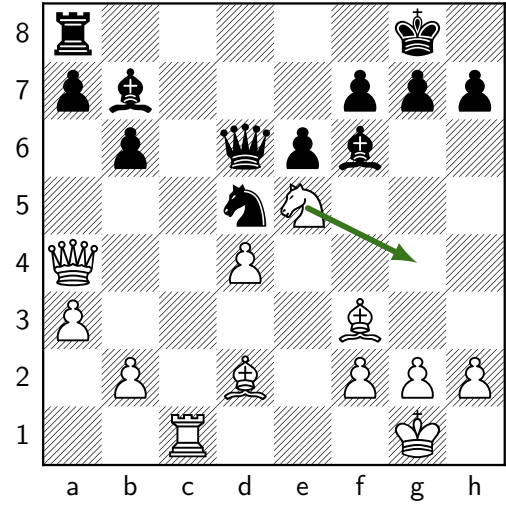
(a)



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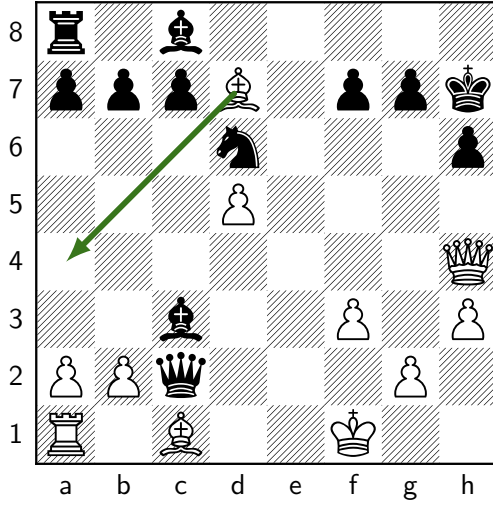


(c)

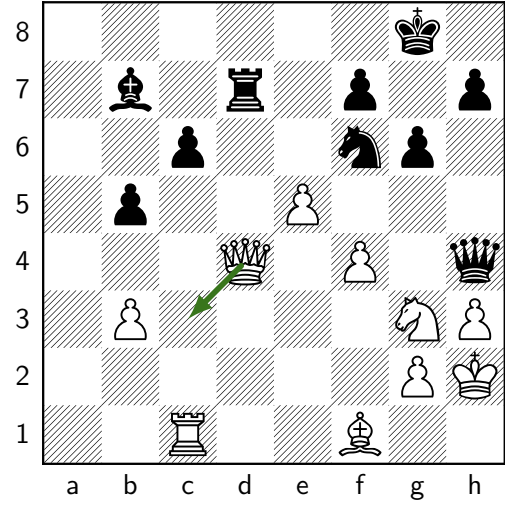


(d)

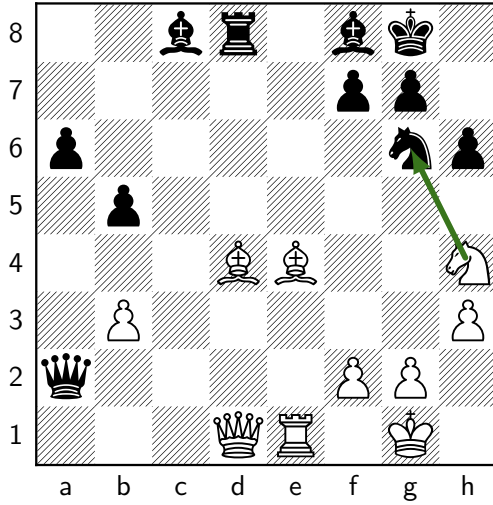
Figure 17: **Move 26, White: Bxd7**. (a) Our position; Melanie plays Bxd7 (g4d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3760` (ply 38) [structural distance $d=118$, QAP obj=99]: the master played e4; under μ our *Ra1* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *Pf3* as their *Rf1*, our *Bg4* as their *Pg2*; transferred move a1b1. (c) Retrieved from `ccr1_top_p0002#2630` (ply 43) [structural distance $d=83$, QAP obj=84]: the master played Ng4; under μ our *Bg4* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Ph3* as their *Rd7*, our *Qh4* as their *Pg7*; transferred move g4h5. (d) Retrieved from `ccr1_top_p0000#3870` (ply 44) [structural distance $d=97$, QAP obj=64]: the master played Ng4; under μ our *Nc3* plays the role of that game’s *Ne5*; cross-type roles: our *nd6* as their *Rc1*, our *Ra1* as their *Pb2*; transferred move c3a4. Synthesis: no retrieved analog transferred to Bxd7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



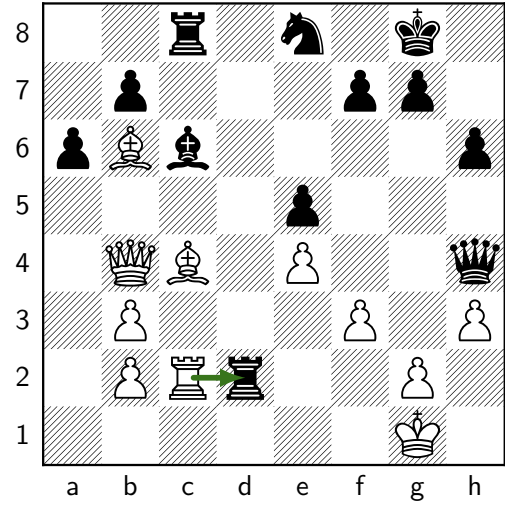
(a)



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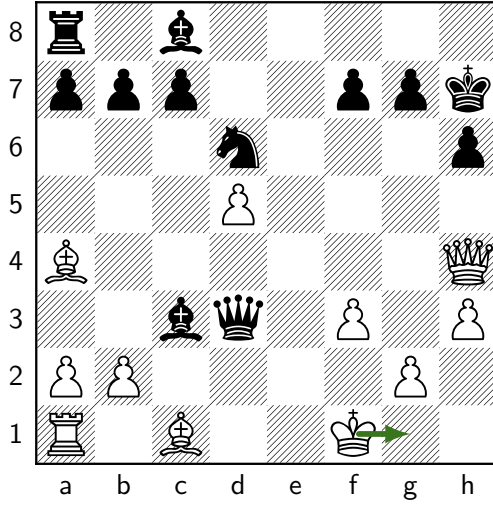


(c)

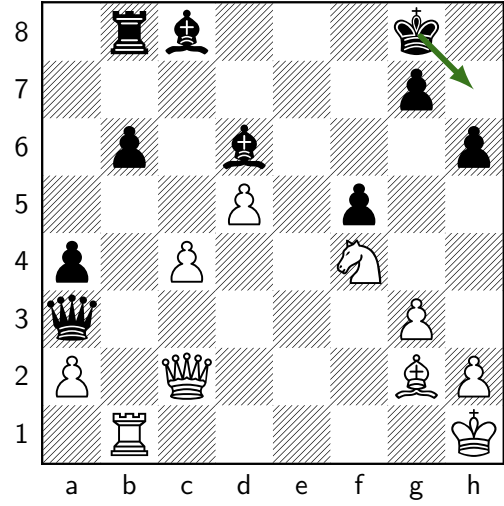


(d)

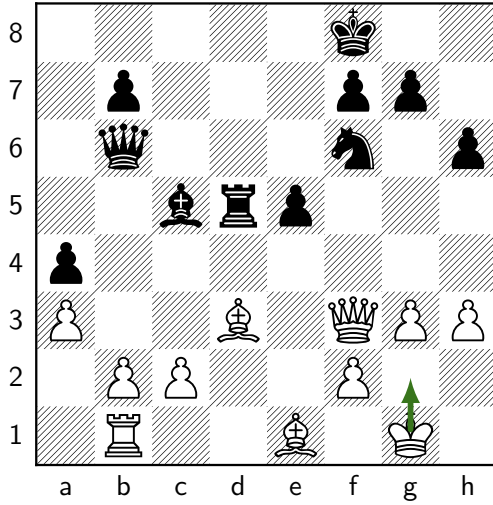
Figure 18: **Move 27, White: Ba4**. (a) Our position; Melanie plays Ba4 (d7a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0000#3091` (ply 60) [structural distance $d=105$, QAP obj=83]: the master played Qc3; under μ our Bd7 plays the role of that game’s Qd4 (a **cross-type** correspondence); cross-type roles: our Pf3 as their Bf1, our Qh4 as their Pb3; transferred move d7a4 (**endorses this move**). (c) Retrieved from `ccr1.top_p0002#2192` (ply 58) [structural distance $d=104$, QAP obj=52]: the master played Nxg6; under μ our Bd7 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our Pf3 as their Qd1, our Pa2 as their Re1; transferred move d7a4 (**endorses this move**). (d) Retrieved from `ccr1.top_p0000#118` (ply 64) [structural distance $d=100$, QAP obj=44]: the master played Rxd2; under μ our Bd7 plays the role of that game’s Rc2 (a **cross-type** correspondence); cross-type roles: our bc8 as their rd2, our Bc1 as their Pf3; transferred move d7a4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ba4 (summed QAP objective 255, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



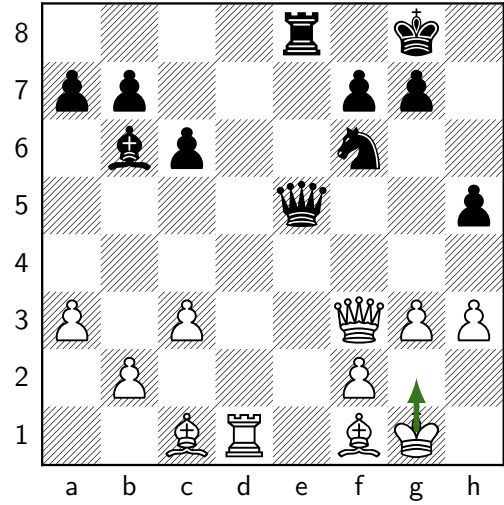
(a)



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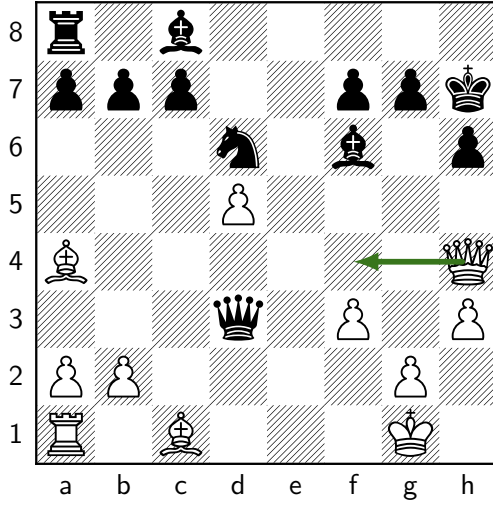


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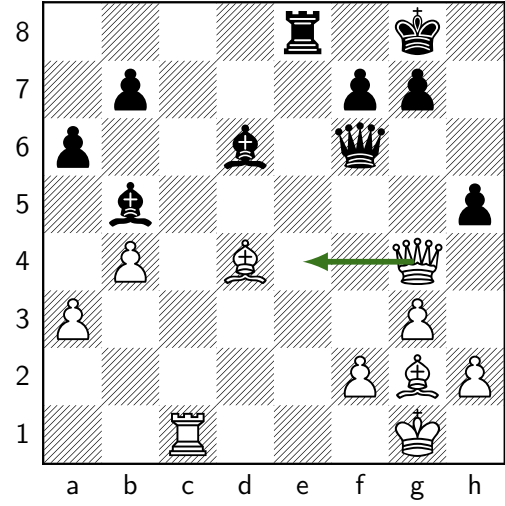


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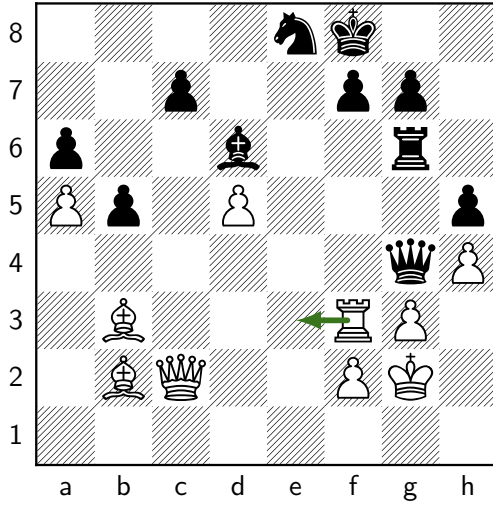
Figure 19: **Move 28, White: Kg1.** (a) Our position; Melanie plays Kg1 ($f1g1$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2783` (ply 65) [structural distance $d=82$, QAP obj=4]: the master played Kh7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Bc1 as their Rb8, our Qh4 as their Bc8; transferred move $f1g1$ (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#983` (ply 66) [structural distance $d=90$, QAP obj=16]: the master played Kg2; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our pa7 as their Be1, our nd6 as their Bd3; transferred move $f1f2$. (d) Retrieved from `ccrl_top_p0000#2969` (ply 50) [structural distance $d=93$, QAP obj=9]: the master played Kg2; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pg2 as their Bc1, our Qh4 as their Rd1; transferred move $f1f2$. Synthesis: 1 retrieved analog transferred to Kg1 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



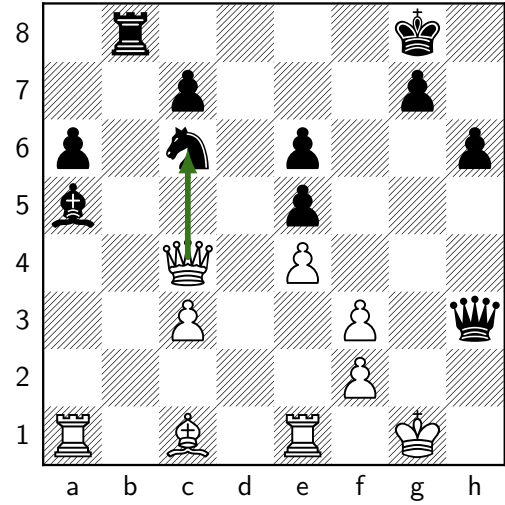
(a)



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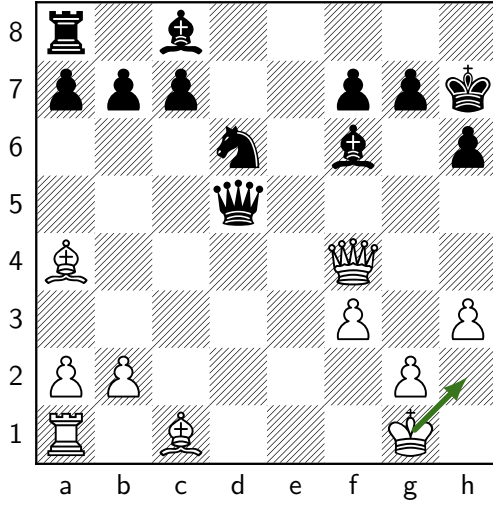


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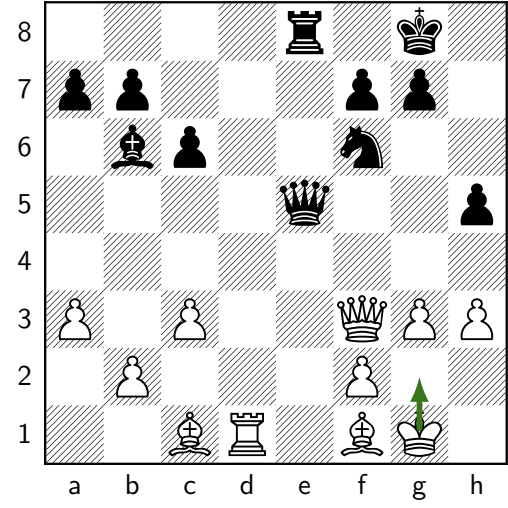


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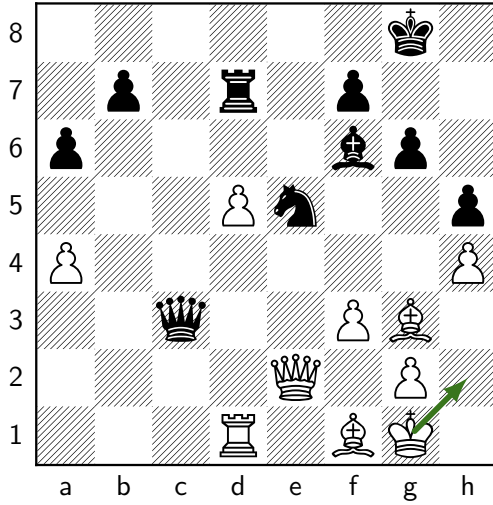
Figure 20: **Move 29, White: Qf4.** (a) Our position; Melanie plays Qf4 (**h4f4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#75` (ply 44) [structural distance $d=94$, QAP obj=56]: the master played Qe4; under μ our Qh4 plays the role of that game’s Qg4; cross-type roles: our Pd5 as their Rc1, our Pf3 as their Bg2; transferred move **h4f4 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#3863` (ply 54) [structural distance $d=93$, QAP obj=51]: the master played Re3; under μ our Qh4 plays the role of that game’s Rf3 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Bb2, our Ba4 as their Qc2; transferred move **h4f4 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#319` (ply 46) [structural distance $d=102$, QAP obj=68]: the master played Qxc6; under μ our Qh4 plays the role of that game’s Qc4; cross-type roles: our bf6 as their Ra1, our pc7 as their Bc1; transferred move **h4f6**. Synthesis: 2 retrieved analogs transferred to Qf4 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



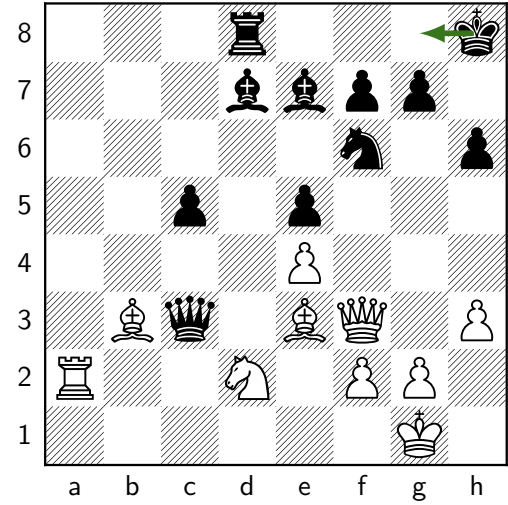
(a)



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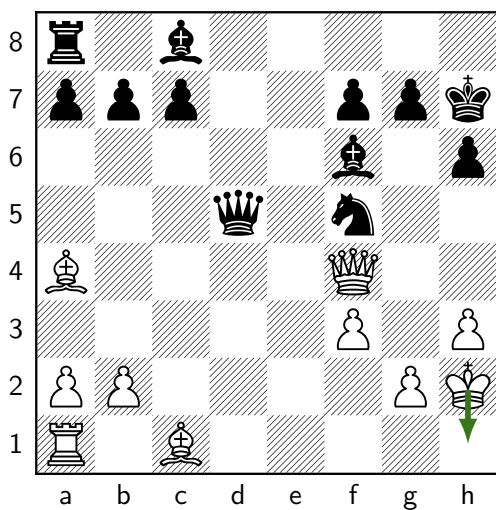


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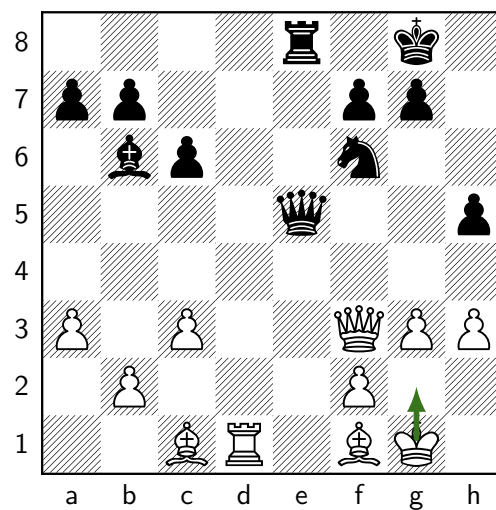


(d)

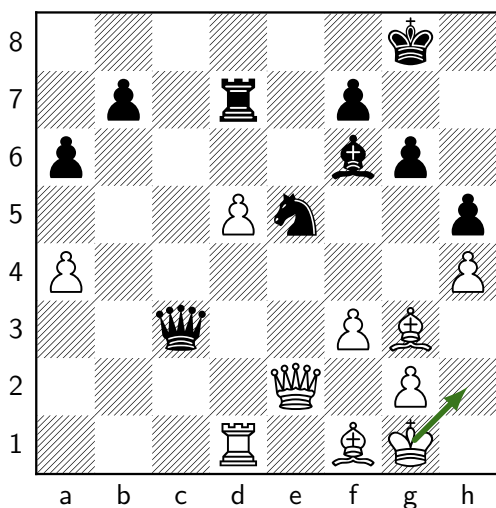
Figure 21: **Move 30, White: Kh2.** (a) Our position; Melanie plays Kh2 (g1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2969` (ply 50) [structural distance $d=98$, QAP obj=72]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Bc1 as their Rd1, our Pg2 as their Bf1; transferred move g1h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1493` (ply 56) [structural distance $d=100$, QAP obj=66]: the master played Kh2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Ba4 as their Rd1, our bf6 as their qc3; transferred move g1h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1812` (ply 53) [structural distance $d=93$, QAP obj=55]: the master played Kg8; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our Ph3 as their Rd8, our qd5 as their Bd7; transferred move g1h2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh2 (summed QAP objective 217, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



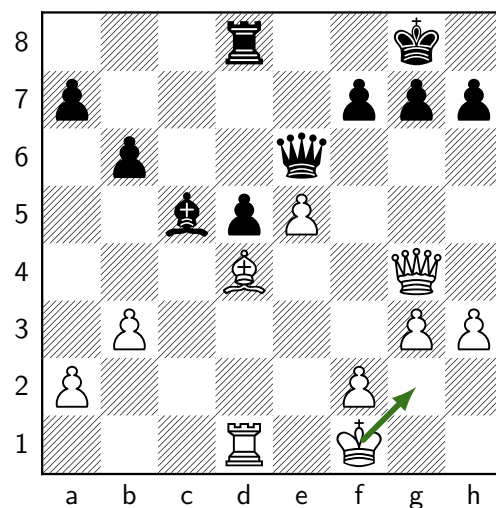
(a)



(b)

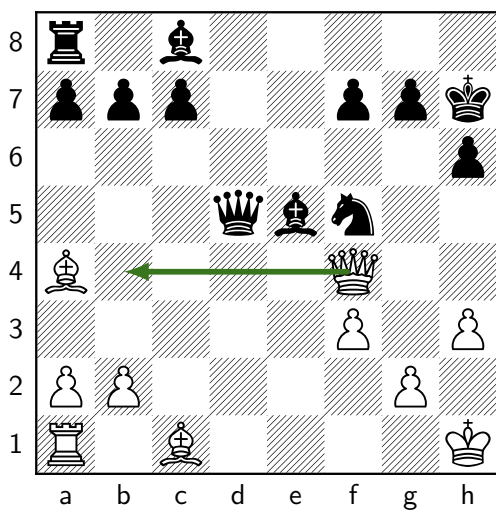


(c)

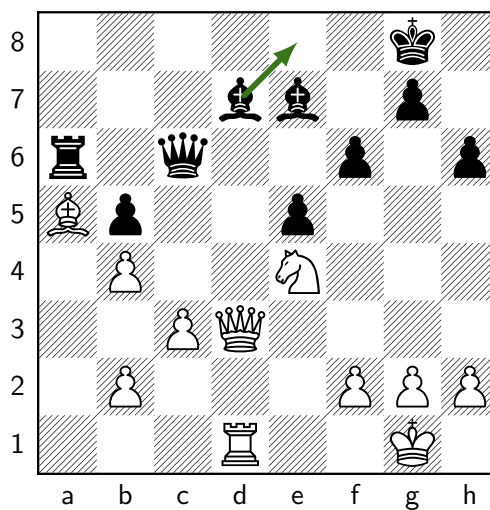


(d)

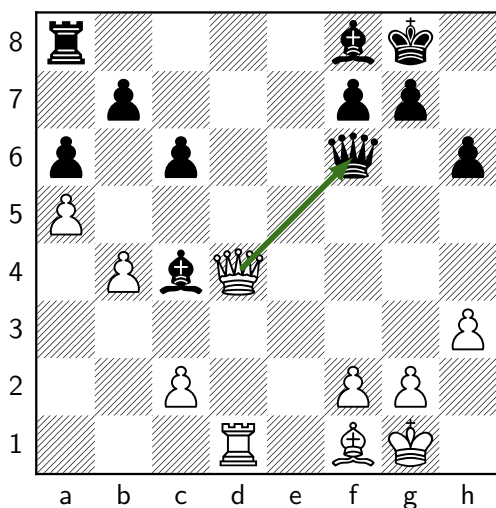
Figure 22: **Move 31, White: Kh1.** (a) Our position; Melanie plays Kh1 (**h2h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2969` (ply 50) [structural distance $d=94$, QAP obj=38]: the master played Kg2; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our qd5 as their Bc1, our pb7 as their Rd1; transferred move h2h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1493` (ply 56) [structural distance $d=95$, QAP obj=31]: the master played Kh2; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our pf7 as their Rd1, our Pg2 as their Bf1; transferred move h2h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2339` (ply 56) [structural distance $d=94$, QAP obj=31]: the master played Kg2; under μ our Kh2 plays the role of that game’s Kf1; cross-type roles: our Ba4 as their Rd1, our Ra1 as their Pf2; transferred move h2h1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kh1 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



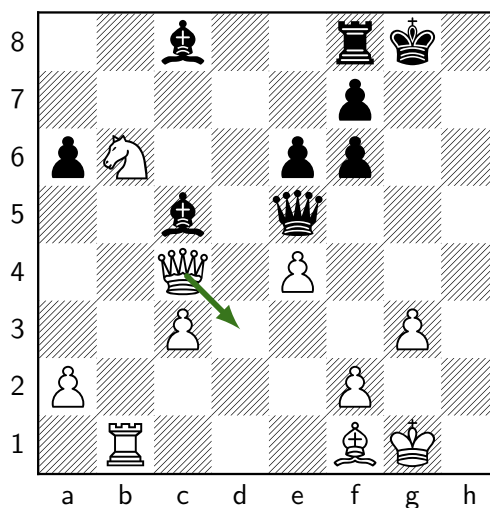
(a)



(b)

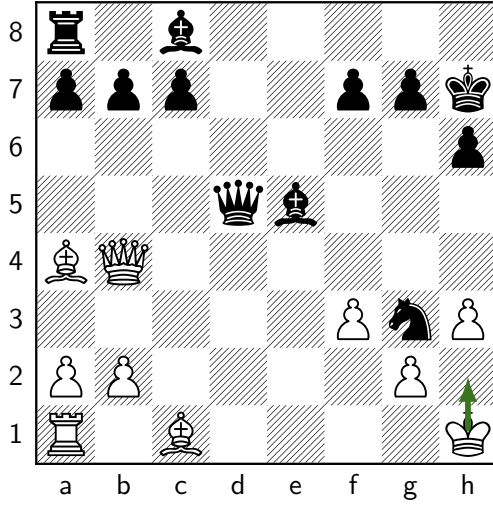


(c)

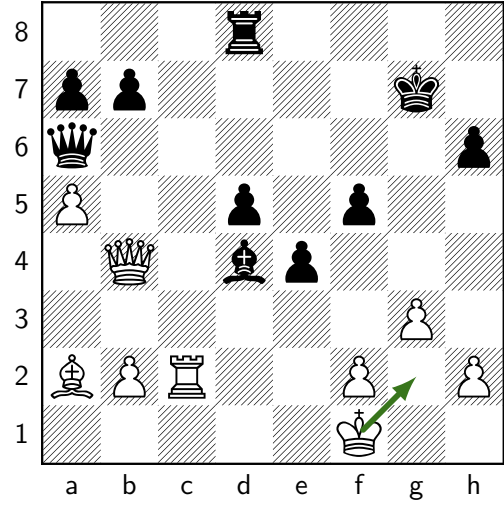


(d)

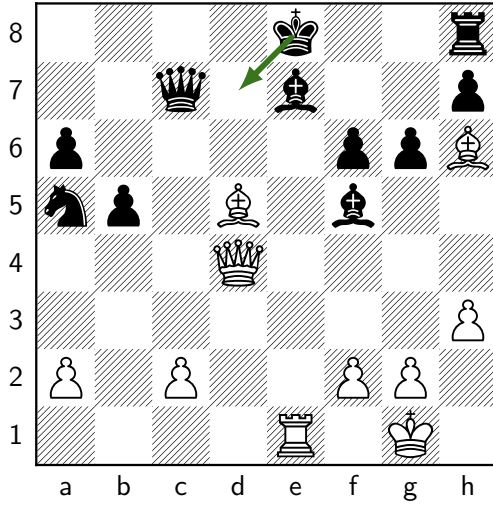
Figure 23: **Move 32, White: Qb4**. (a) Our position; Melanie plays **Qb4** (**f4b4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3660` (ply 51) [structural distance $d=106$, QAP obj=27]: the master played **Be8**; under μ our *Qf4* plays the role of that game’s *Bd2* (a **cross-type** correspondence); cross-type roles: our *qd5* as their *Be7*, our *Qf4* as their *Ra6*; transferred move **f4b4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2678` (ply 46) [structural distance $d=109$, QAP obj=66]: the master played **Qxf6**; under μ our *Qf4* plays the role of that game’s *Qd4*; cross-type roles: our *Pf3* as their *Rd1*, our *pf7* as their *Bf1*; transferred move **f4f5**. (d) Retrieved from `ccr1_top_p0002#2220` (ply 50) [structural distance $d=118$, QAP obj=58]: the master played **Qd3**; under μ our *Qf4* plays the role of that game’s *Qc4*; cross-type roles: our *Ba4* as their *Pe4*, our *nf5* as their *pa6*; transferred move **f4e3**. Synthesis: 1 retrieved analog transferred to **Qb4** (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



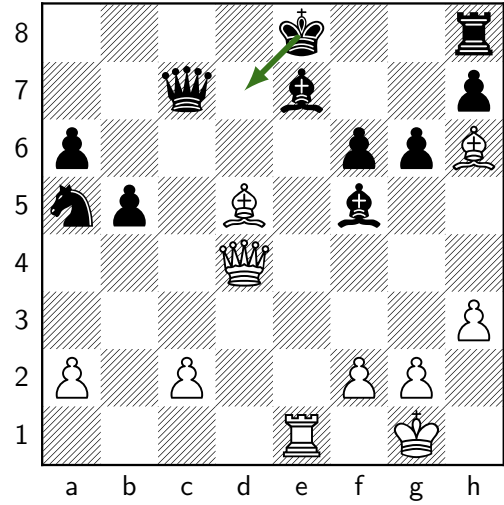
(a)



(b)

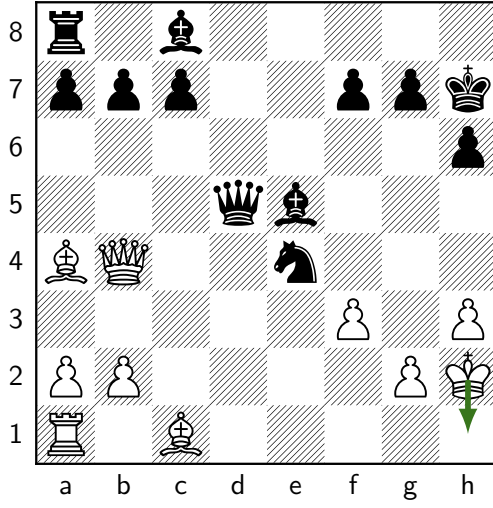


(c)

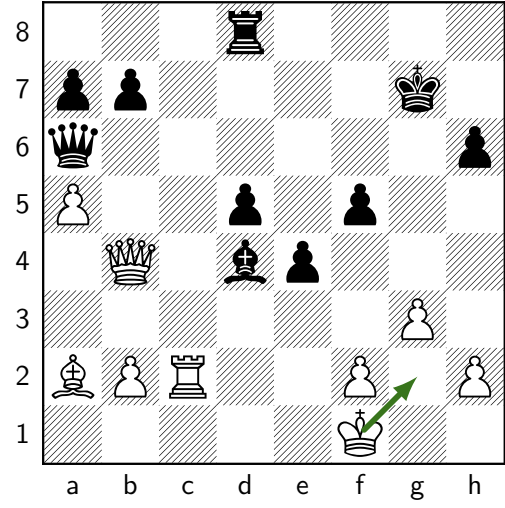


(d)

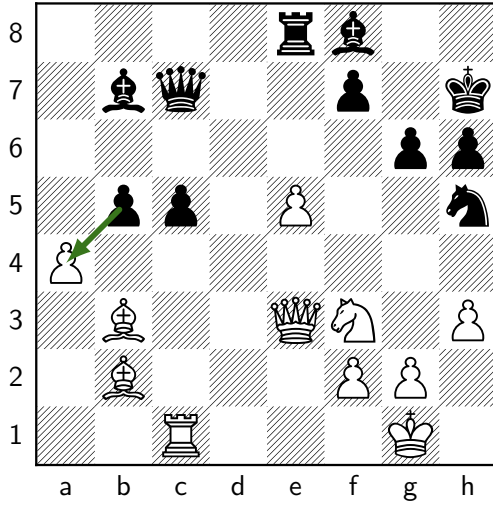
Figure 24: **Move 33, White: Kh2.** (a) Our position; Melanie plays Kh2 (**h1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#272` (ply 64) [structural distance $d=86$, QAP obj=31]: the master played Kg2; under μ our Kh1 plays the role of that game’s Kf1; cross-type roles: our Ph3 as their Ba2, our Ba4 as their Rc2; transferred move **h1h2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#728` (ply 53) [structural distance $d=91$, QAP obj=23]: the master played Kd7; under μ our Kh1 plays the role of that game’s Ke1; cross-type roles: our Pb2 as their Be7, our Ba4 as their Pg6; transferred move **h1g1**. (d) Retrieved from `ccr1_top_p0002#2255` (ply 53) [structural distance $d=91$, QAP obj=23]: the master played Kd7; under μ our Kh1 plays the role of that game’s Ke1; cross-type roles: our Pb2 as their Be7, our Ba4 as their Pg6; transferred move **h1g1**. Synthesis: 1 retrieved analog transferred to Kh2 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



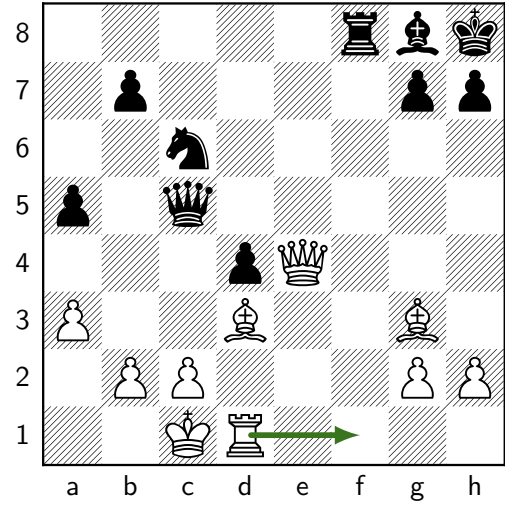
(a)



(b)

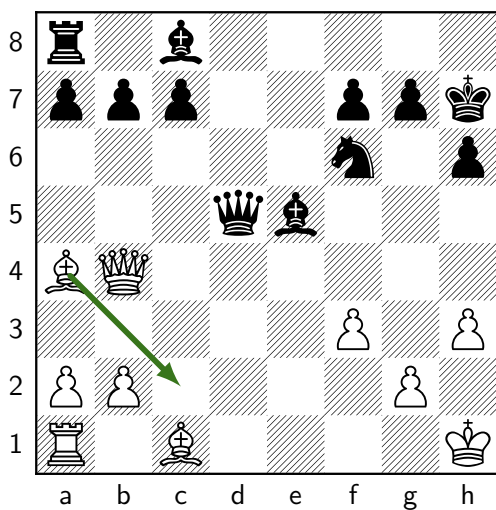


(c)

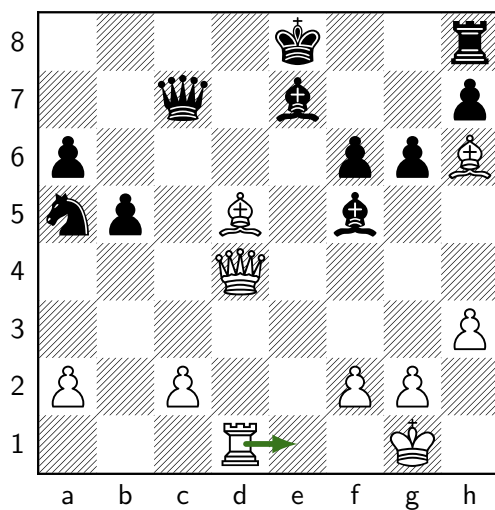


(d)

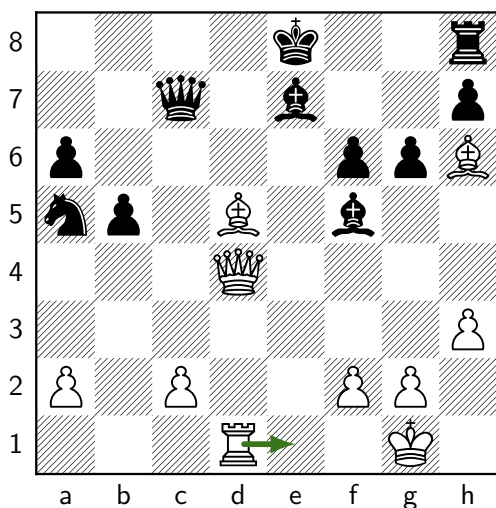
Figure 25: **Move 34, White: Kh1.** (a) Our position; Melanie plays Kh1 (**h2h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#272` (ply 64) [structural distance $d=93$, QAP obj=55]: the master played Kg2; under μ our Kh2 plays the role of that game’s Kf1; cross-type roles: our Pa2 as their Ba2, our Ba4 as their Pb2; transferred move h2h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#310` (ply 59) [structural distance $d=95$, QAP obj=34]: the master played bxa4; under μ our Bc1 plays the role of that game’s Pb4 (a **cross-type** correspondence); cross-type roles: our be5 as their Re8, our pc7 as their Bf8; transferred move c1f4. (d) Retrieved from `ccr1_top_p0000#52` (ply 50) [structural distance $d=93$, QAP obj=26]: the master played Rf1; under μ our Pg2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Ba4 as their Pb2, our ne4 as their Bd3; transferred move g2g3. Synthesis: 1 retrieved analog transferred to Kh1 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



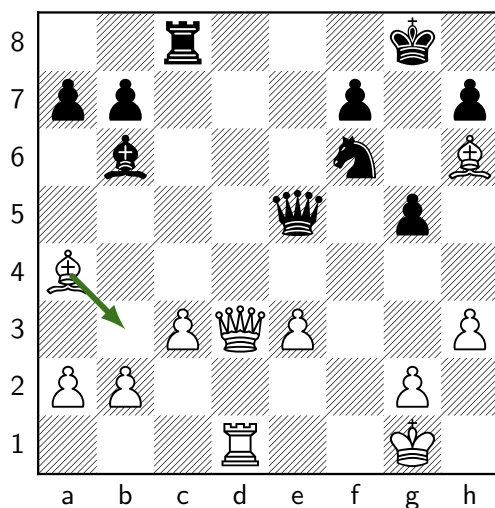
(a)



(b)

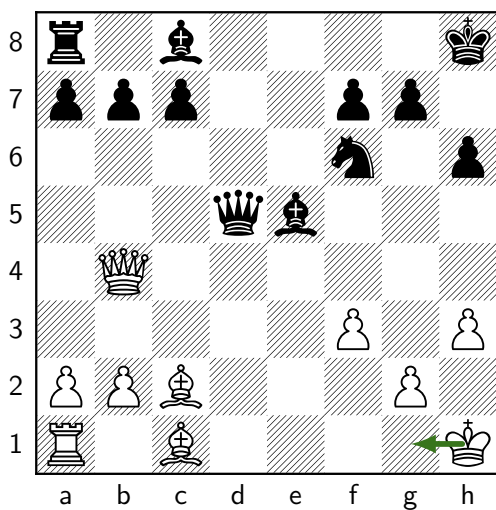


(c)

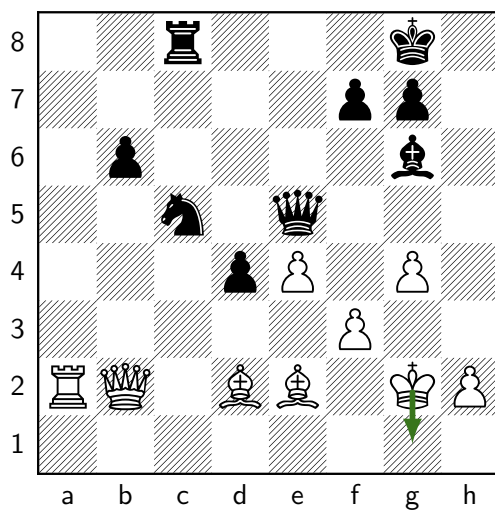


(d)

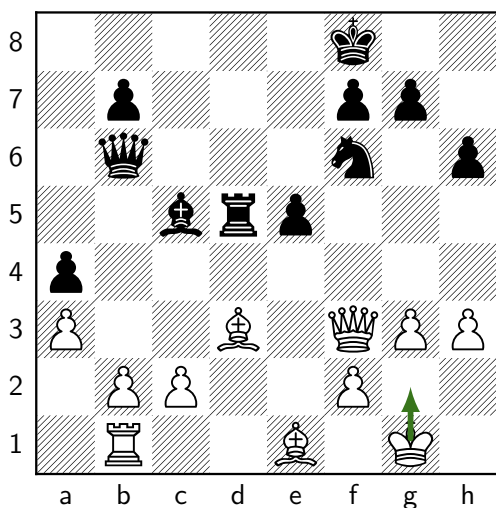
Figure 26: **Move 35, White: Bc2+.** (a) Our position; Melanie plays Bc2+ (a4c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#728` (ply 52) [structural distance $d=82$, QAP obj=63]: the master played **Re1**; under μ our *Ba4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *be5* as their *Qd4*, our *Qb4* as their *na5*; transferred move **a4c2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2255` (ply 52) [structural distance $d=82$, QAP obj=63]: the master played **Re1**; under μ our *Ba4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *be5* as their *Qd4*, our *Qb4* as their *na5*; transferred move **a4c2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3489` (ply 54) [structural distance $d=70$, QAP obj=53]: the master played **Bb3**; under μ our *Ba4* plays the role of that game’s *Ba4*; cross-type roles: our *bc8* as their *Ph3*, our *be5* as their *qe5*; transferred move **a4c2** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc2+ (summed QAP objective 205, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



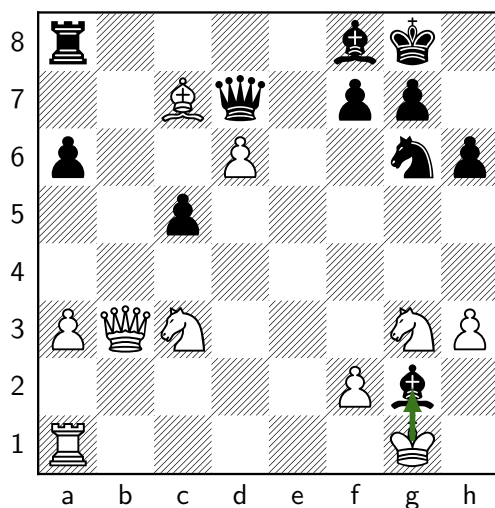
(a)



(b)

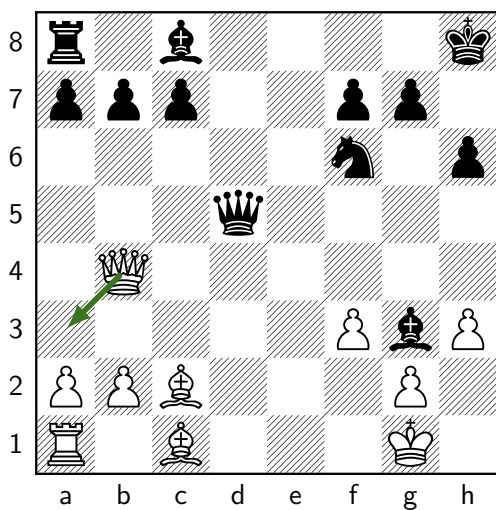


(c)

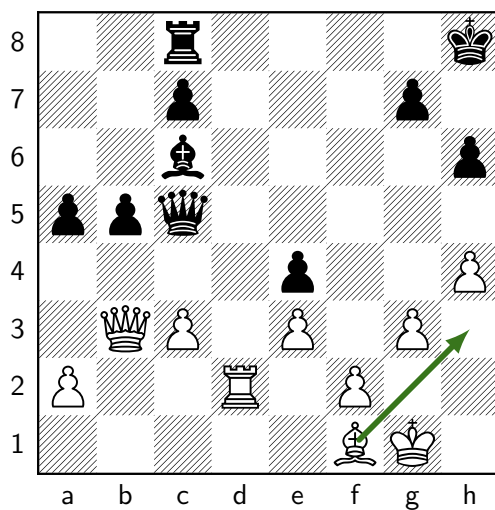


(d)

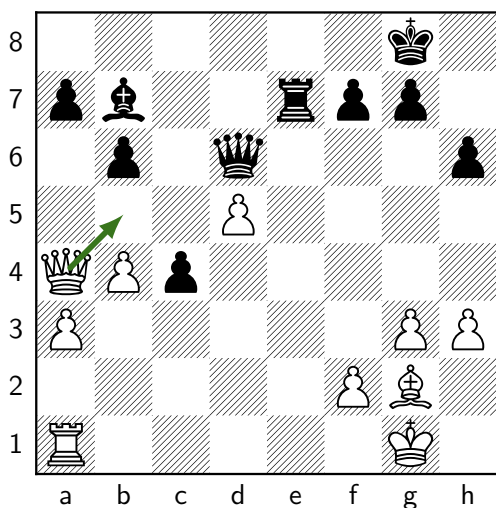
Figure 27: **Move 36, White: Kg1 (h1g1).** (a) Our position; Melanie plays Kg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2209` (ply 86) [structural distance $d=78$, QAP obj=42]: the master played Kg1; under μ our *Kh1* plays the role of that game’s *Kg2*; cross-type roles: our *nf6* as their *Ra2*, our *be5* as their *Qb2*; transferred move h1g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#983` (ply 66) [structural distance $d=69$, QAP obj=35]: the master played Kg2; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *Bc1* as their *Rb1*, our *Pa2* as their *Be1*; transferred move h1g1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#348` (ply 54) [structural distance $d=77$, QAP obj=21]: the master played Kxg2; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *Pa2* as their *Ra1*, our *Pg2* as their *bg2*; transferred move h1g1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kg1 (summed QAP objective 148, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



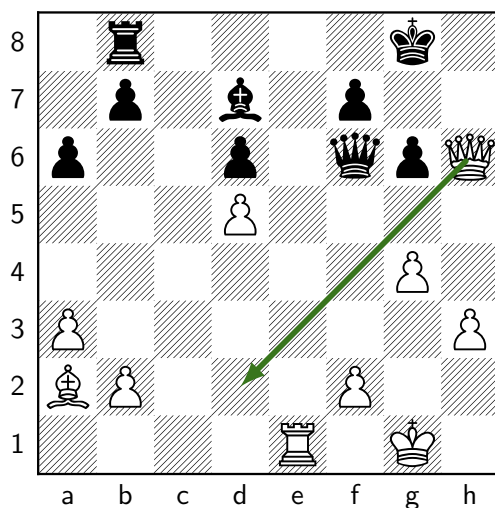
(a)



(b)

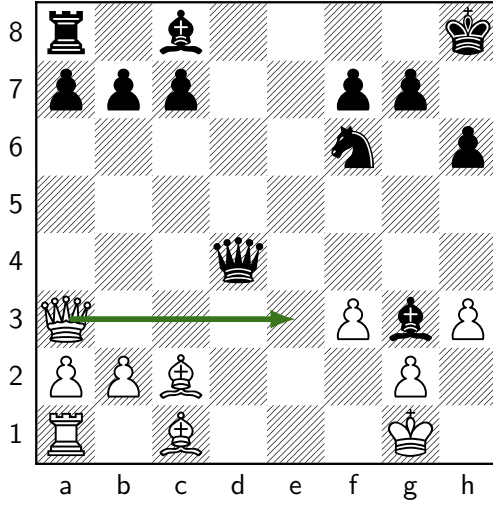


(c)

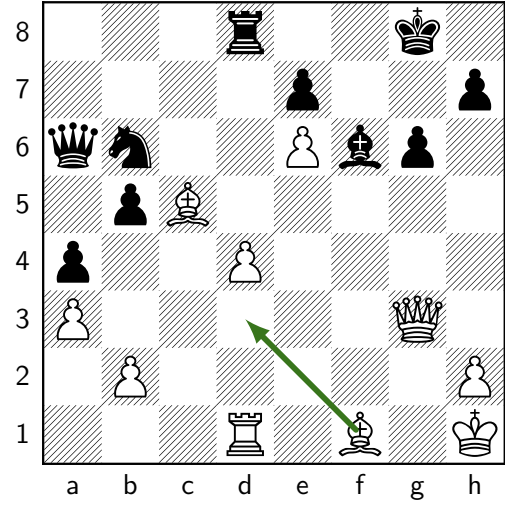


(d)

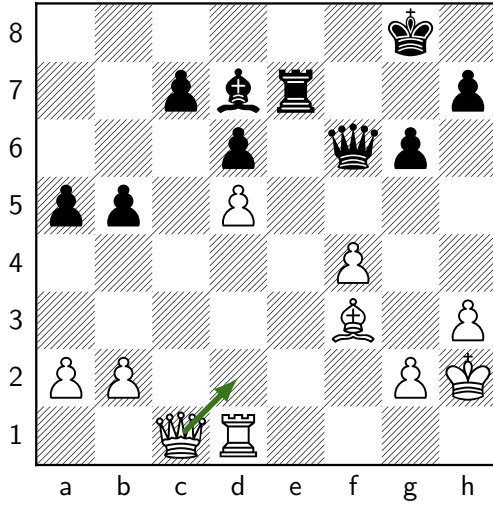
Figure 28: **Move 37, White: Qa3.** (a) Our position; Melanie plays Qa3 (b4a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1374` (ply 76) [structural distance $d=70$, QAP obj=58]: the master played Bh3; under μ our Qb4 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Rd2, our bg3 as their pe4; transferred move b4a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3475` (ply 56) [structural distance $d=71$, QAP obj=37]: the master played Qb5; under μ our Qb4 plays the role of that game’s Qa4; cross-type roles: our Bc2 as their Pf2, our Ra1 as their Bg2; transferred move b4a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#189` (ply 84) [structural distance $d=77$, QAP obj=26]: the master played Qd2; under μ our Qb4 plays the role of that game’s Qh6; cross-type roles: our Pg2 as their Ba2, our bg3 as their Pa3; transferred move b4a3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qa3 (summed QAP objective 177, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



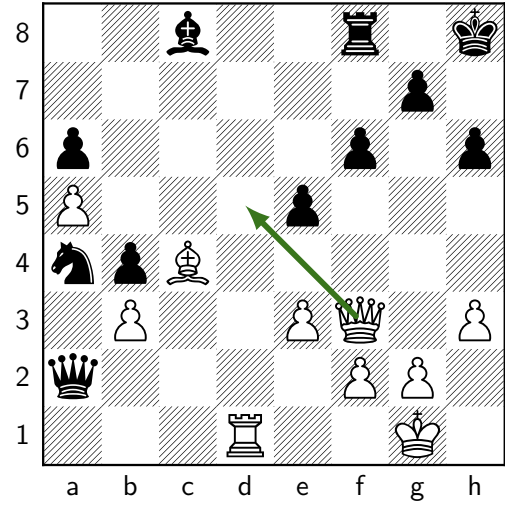
(a)



(b)

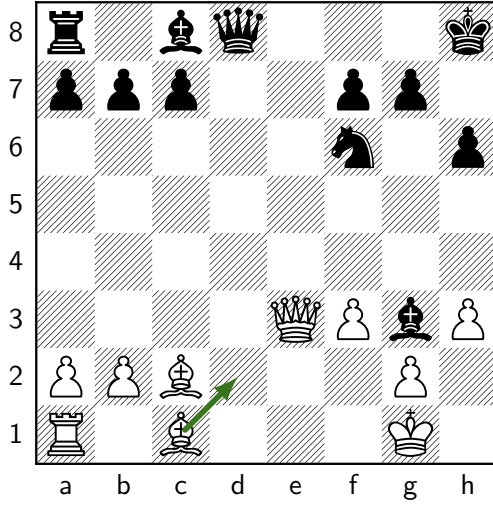


(c)

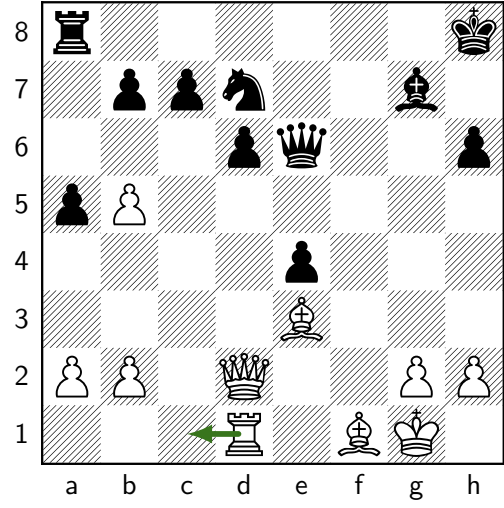


(d)

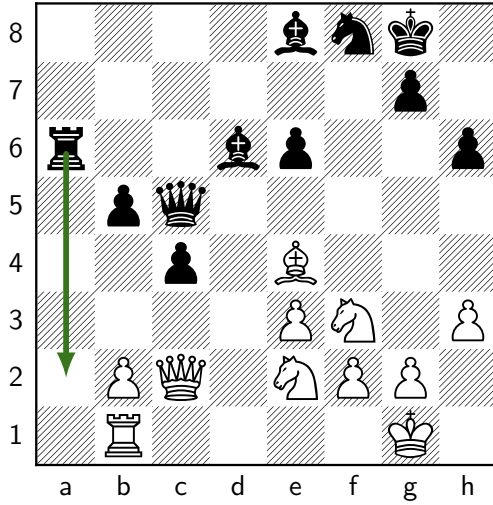
Figure 29: **Move 38, White: Qe3.** (a) Our position; Melanie plays Qe3 (a3e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#81` (ply 78) [structural distance $d=79$, QAP obj=32]: the master played Bd3; under μ our Qa3 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Rd1, our nf6 as their Pa3; transferred move a3e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1287` (ply 62) [structural distance $d=72$, QAP obj=26]: the master played Qd2; under μ our Qa3 plays the role of that game’s Qc1; cross-type roles: our Pa2 as their Rd1, our Ra1 as their Pa2; transferred move a3e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#833` (ply 66) [structural distance $d=70$, QAP obj=21]: the master played Qd5; under μ our Qa3 plays the role of that game’s Qf3; cross-type roles: our Bc2 as their Rd1, our nf6 as their Pe3; transferred move a3e3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qe3 (summed QAP objective 169, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



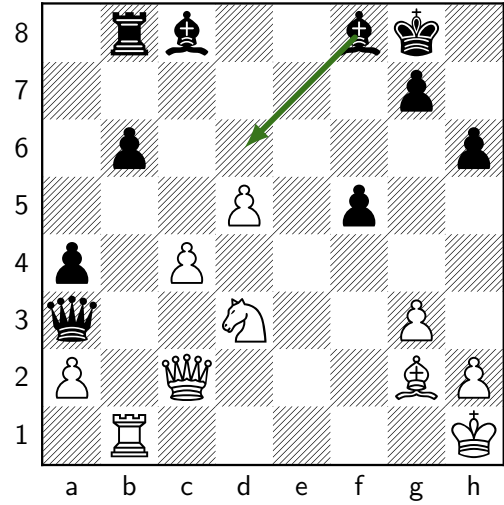
(a)



(b)

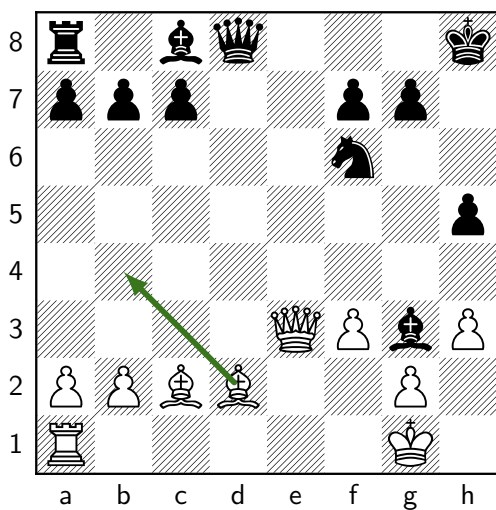


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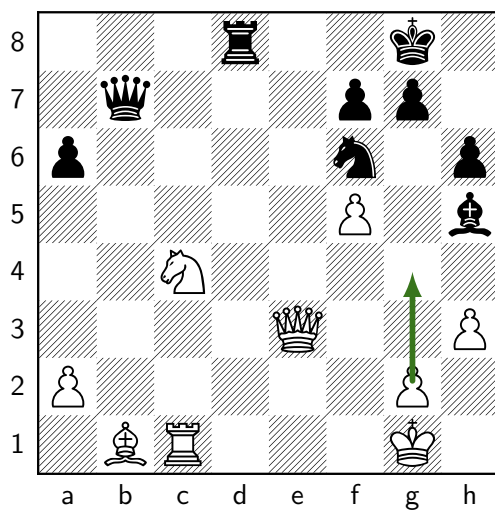


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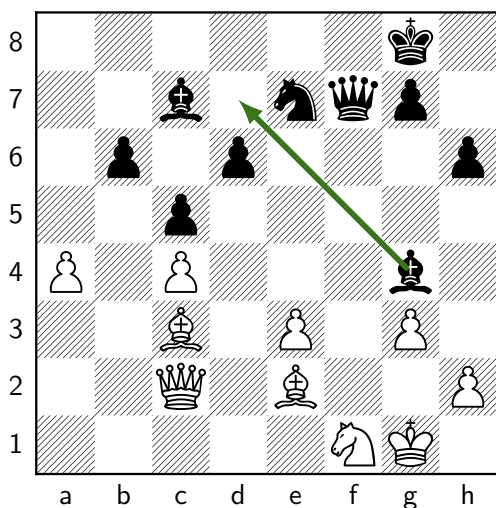
Figure 30: **Move 39, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2581` (ply 62) [structural distance $d=73$, QAP obj=64]: the master played Rc1; under μ our Bc1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Bf1, our Bc2 as their Pg2; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1992` (ply 67) [structural distance $d=68$, QAP obj=22]: the master played Ra2; under μ our Bc1 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Be8, our pc7 as their Nf8; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2783` (ply 63) [structural distance $d=58$, QAP obj=14]: the master played Bd6; under μ our Bc1 plays the role of that game’s Bf1; cross-type roles: our Pg2 as their Rb8, our Ra1 as their Bc8; transferred move c1d2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bd2 (summed QAP objective 148, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



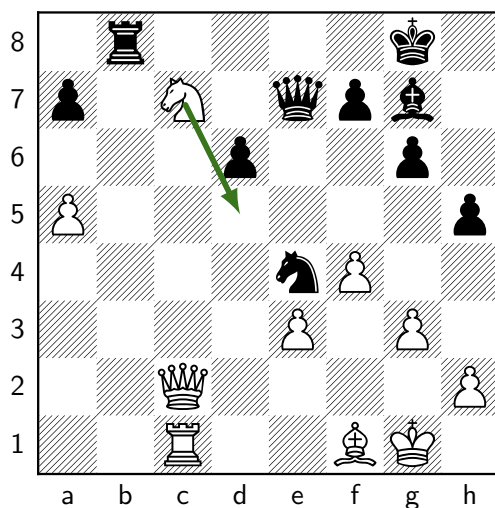
(a)



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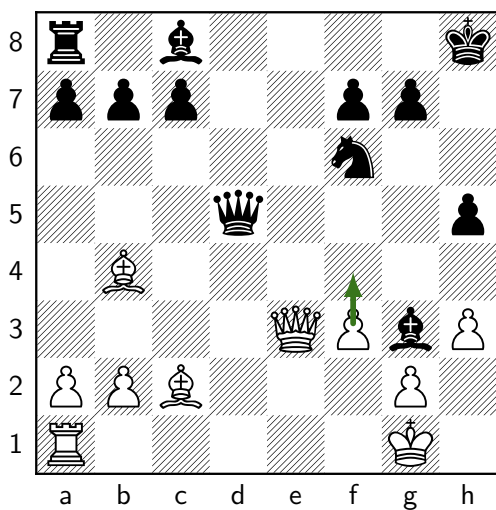


(c)

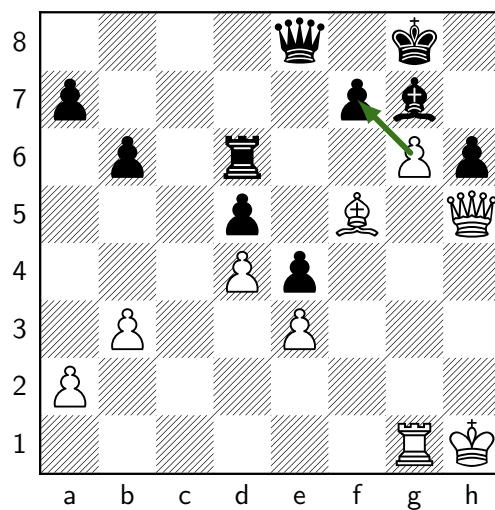


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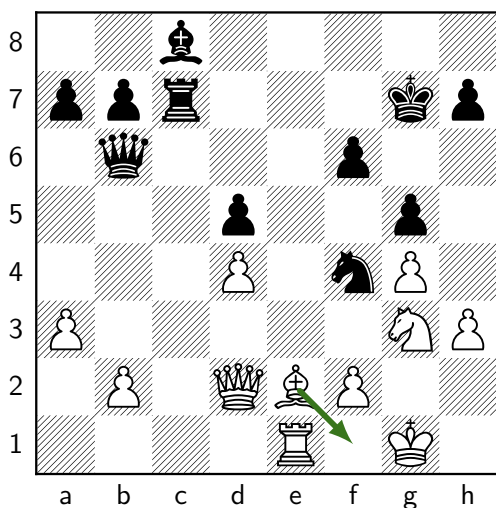
Figure 31: **Move 40, White: Bb4**. (a) Our position; Melanie plays Bb4 (d2b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#975` (ply 64) [structural distance $d=73$, QAP obj=39]: the master played g4; under μ our Bd2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Bc2 as their Pa2, our Pg2 as their Nc4; transferred move d2b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2649` (ply 71) [structural distance $d=77$, QAP obj=12]: the master played Bd7; under μ our Bd2 plays the role of that game’s Bg5; cross-type roles: our Pa2 as their Bc7, our Bc2 as their Ne7; transferred move d2b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2125` (ply 80) [structural distance $d=82$, QAP obj=11]: the master played Nd5; under μ our Bd2 plays the role of that game’s Nc7 (a **cross-type** correspondence); cross-type roles: our Bc2 as their Rc1, our bc8 as their Pg3; transferred move d2b4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bb4 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



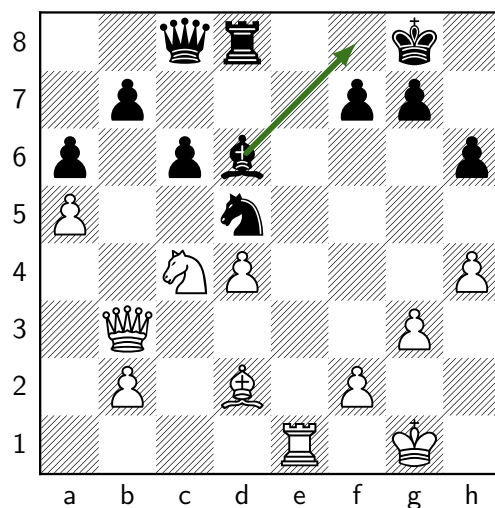
(a)



(b)

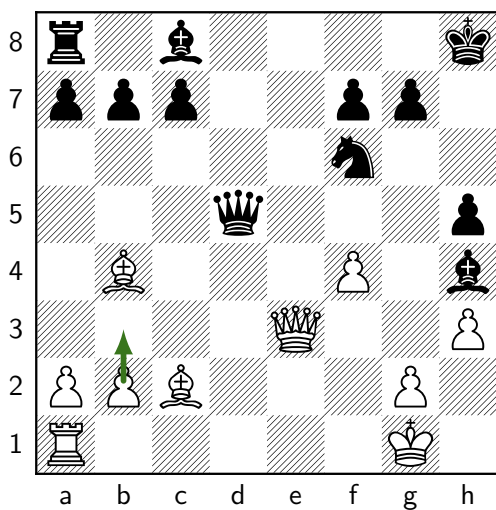


(c)

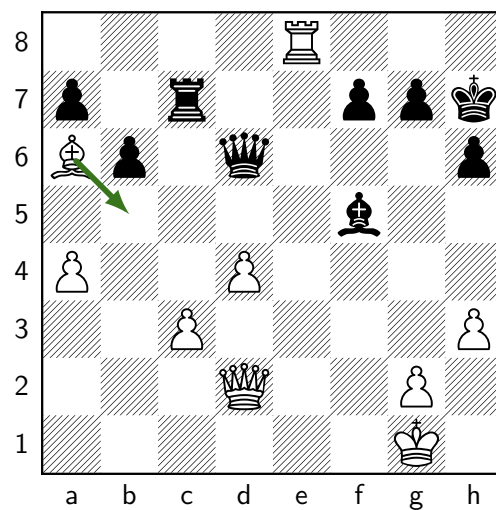


(d)

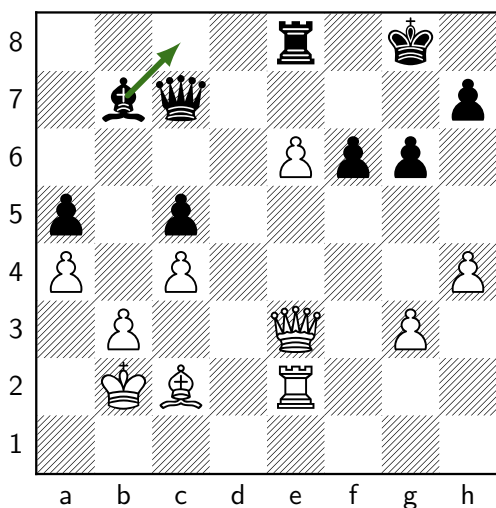
Figure 32: **Move 41, White: f4.** (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2103` (ply 68) [structural distance $d=93$, QAP obj=23]: the master played `gxf7+`; under μ our *Pf3* plays the role of that game's *Pg6*; cross-type roles: our *bg3* as their *Rg1*, our *bc8* as their *Pa2*; transferred move `f3f4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2963` (ply 72) [structural distance $d=90$, QAP obj=17]: the master played `Bf1`; under μ our *Pf3* plays the role of that game's *Be2* (a **cross-type** correspondence); cross-type roles: our *Bc2* as their *Qd2*, our *Qe3* as their *Ng3*; transferred move `f3f4` (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2433` (ply 65) [structural distance $d=83$, QAP obj=12]: the master played `Bf8`; under μ our *Pf3* plays the role of that game's *Bd3* (a **cross-type** correspondence); cross-type roles: our *Ra1* as their *Qc8*, our *Pb2* as their *Rd8*; transferred move `f3f4` (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f4 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



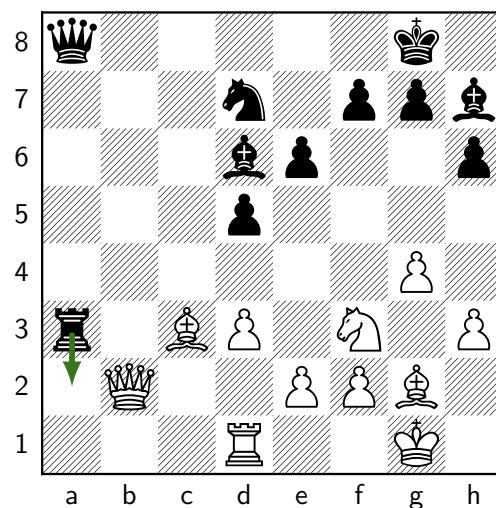
(a)



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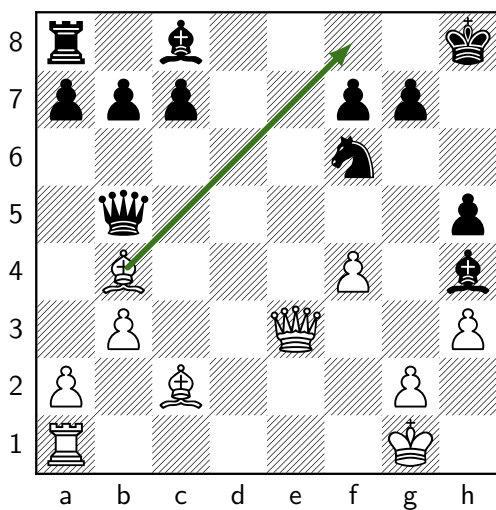


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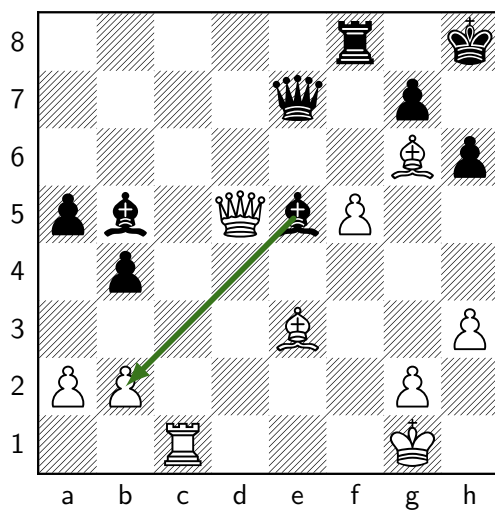


(d)

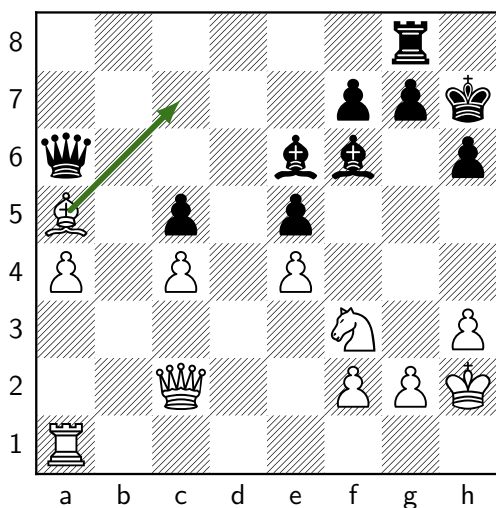
Figure 33: **Move 42, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3229` (ply 66) [structural distance $d=101$, QAP obj=41]: the master played Bb5; under μ our *Pb2* plays the role of that game’s *Ba6* (a **cross-type** correspondence); cross-type roles: our *bh4* as their *Pa4*, our *qd5* as their *bf5*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3814` (ply 73) [structural distance $d=100$, QAP obj=41]: the master played Bc8; under μ our *Pb2* plays the role of that game’s *Bb2* (a **cross-type** correspondence); cross-type roles: our *Qe3* as their *Re8*, our *bc8* as their *Qc7*; transferred move b2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#919` (ply 73) [structural distance $d=103$, QAP obj=4]: the master played Ra2; under μ our *Pb2* plays the role of that game’s *Ra6* (a **cross-type** correspondence); cross-type roles: our *Pa2* as their *Qa8*, our *Pg2* as their *Nd7*; transferred move b2b3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to b3 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



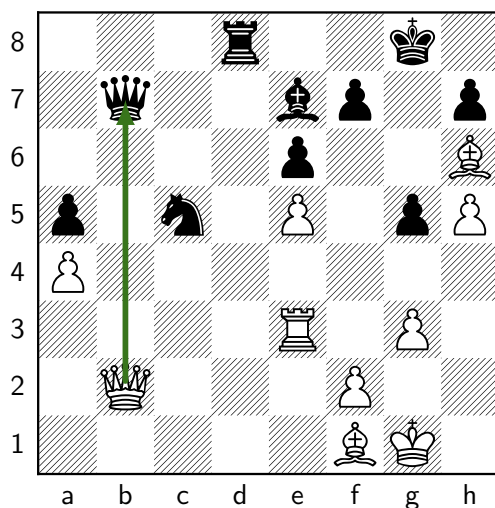
(a)



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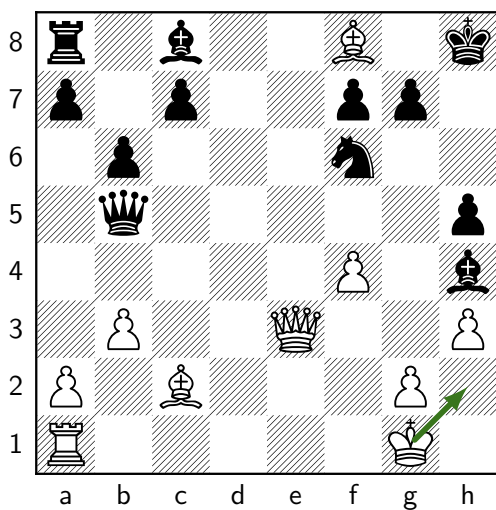


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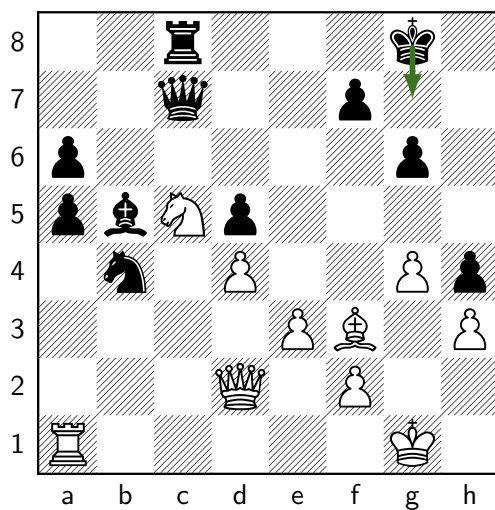


(d)

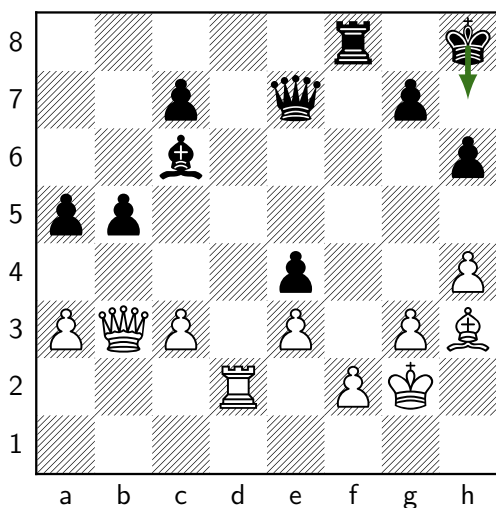
Figure 34: **Move 43, White: Bf8.** (a) Our position; Melanie plays Bf8 (b4f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#410` (ply 75) [structural distance $d=95$, QAP obj=31]: the master played Bxb2; under μ our Bb4 plays the role of that game’s Be4; cross-type roles: our Qe3 as their Rf8, our Pb3 as their Qe7; transferred move b4f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1282` (ply 86) [structural distance $d=90$, QAP obj=25]: the master played Bc7; under μ our Bb4 plays the role of that game’s Ba5; cross-type roles: our Bc2 as their Ra1, our Pa2 as their Qc2; transferred move b4f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2120` (ply 80) [structural distance $d=95$, QAP obj=15]: the master played Qxb7; under μ our Bb4 plays the role of that game’s Qb2 (a **cross-type** correspondence); cross-type roles: our Pf4 as their Bf1, our Qe3 as their Re3; transferred move b4f8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bf8 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



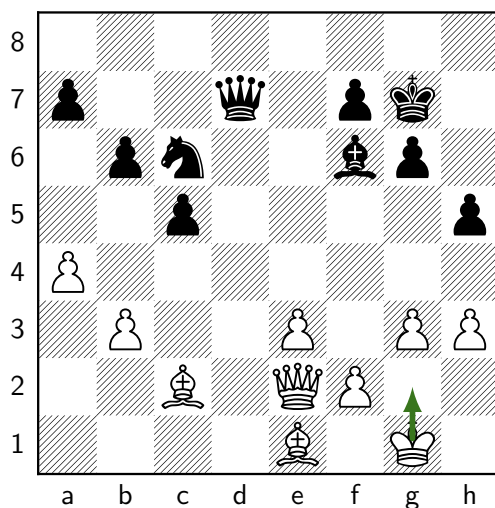
(a)



(b)

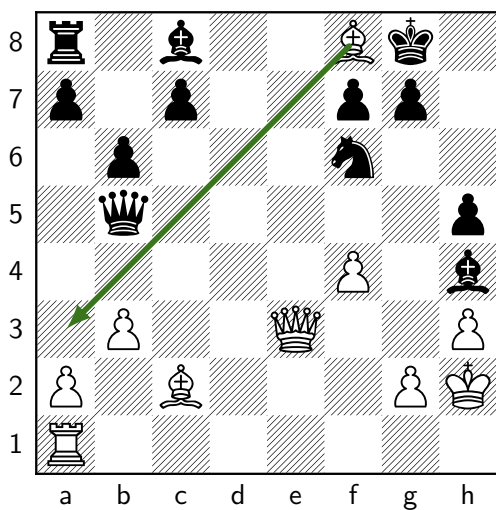


(c)

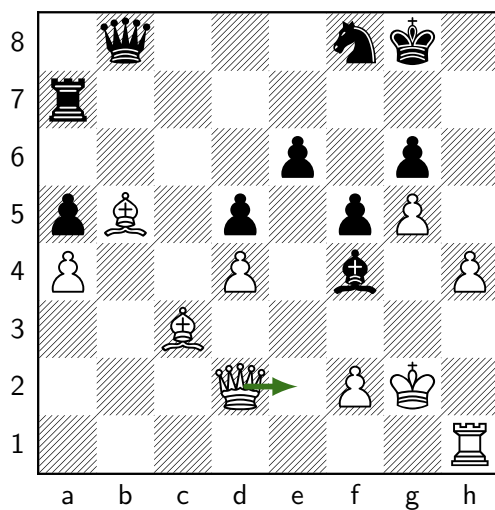


(d)

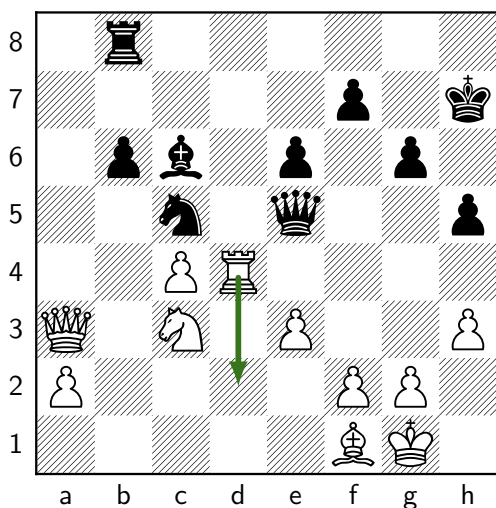
Figure 35: **Move 44, White: Kh2.** (a) Our position; Melanie plays Kh2 (g1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#933` (ply 81) [structural distance $d=93$, QAP obj=26]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our bc8 as their Pg6, our Pb3 as their nc5; transferred move g1h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1374` (ply 81) [structural distance $d=92$, QAP obj=24]: the master played Kh7; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our Bf8 as their Rf8, our bh4 as their Ph6; transferred move g1h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#472` (ply 72) [structural distance $d=93$, QAP obj=20]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Ph3 as their Be1, our Ra1 as their Pf2; transferred move g1h2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh2 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



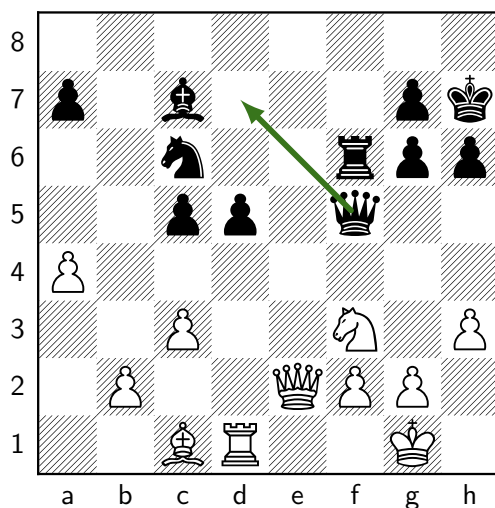
(a)



(b)



(c)



(d)

Figure 36: **Move 45, White: Ba3**. (a) Our position; Melanie plays Ba3 (f8a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#748` (ply 104) [structural distance $d=74$, QAP obj=12]: the master played Qe2; under μ our Bf8 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Rh1, our kg8 as their bf4; transferred move f8a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#262` (ply 78) [structural distance $d=86$, QAP obj=10]: the master played Rd2; under μ our Bf8 plays the role of that game’s Rd4 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Bf1, our Qe3 as their Nc3; transferred move f8a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3570` (ply 97) [structural distance $d=83$, QAP obj=9]: the master played Qd7; under μ our Bf8 plays the role of that game’s Qf4 (a **cross-type** correspondence); cross-type roles: our qb5 as their Bc7, our ra8 as their Nc6; transferred move f8a3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ba3 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

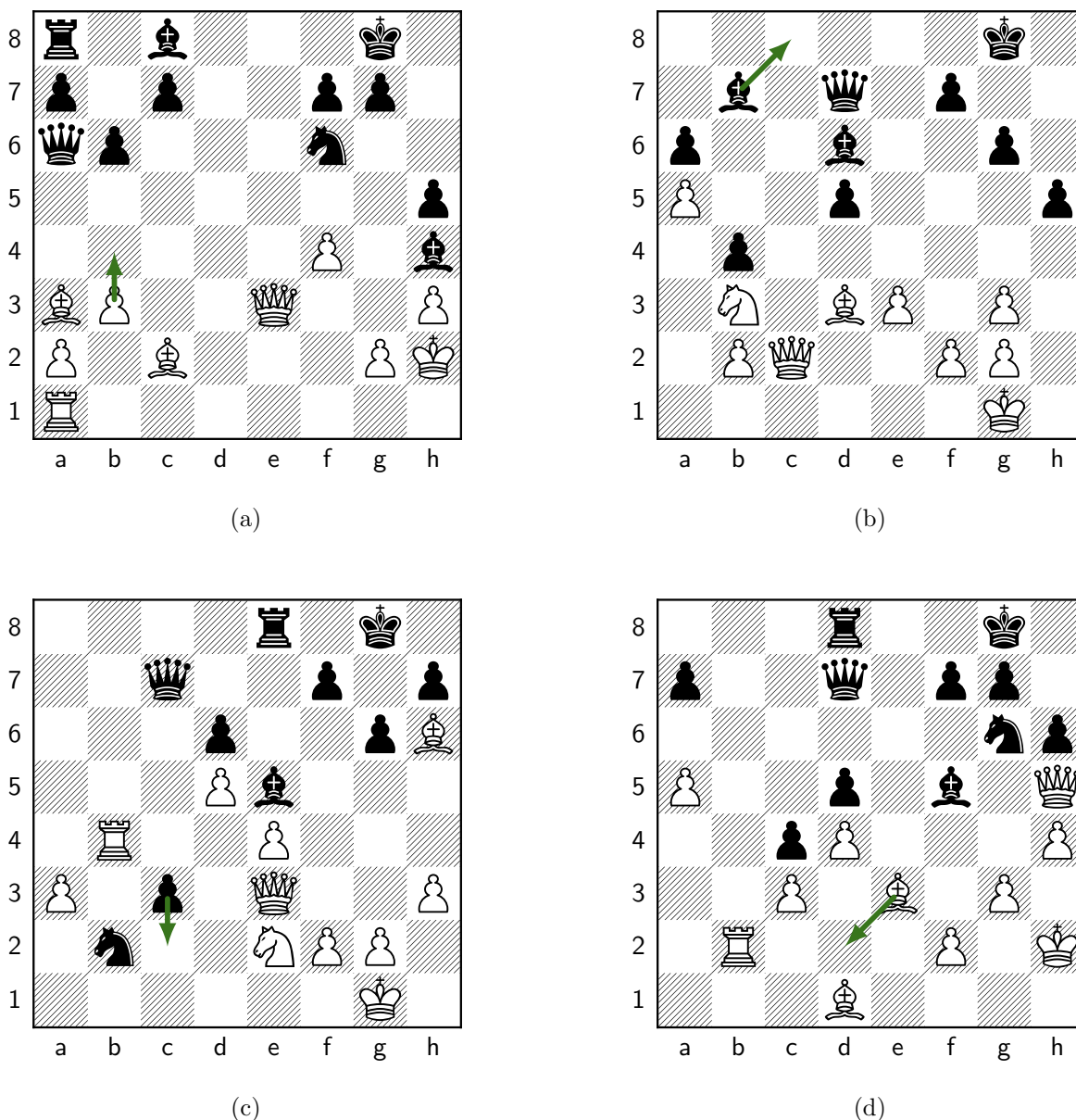
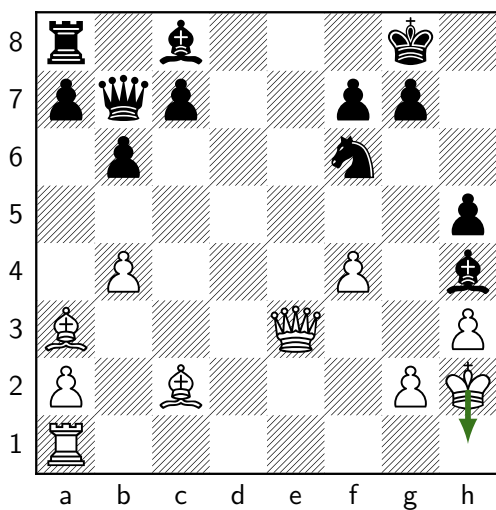
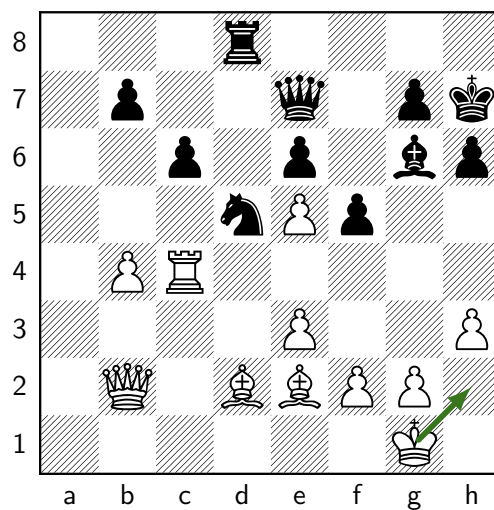


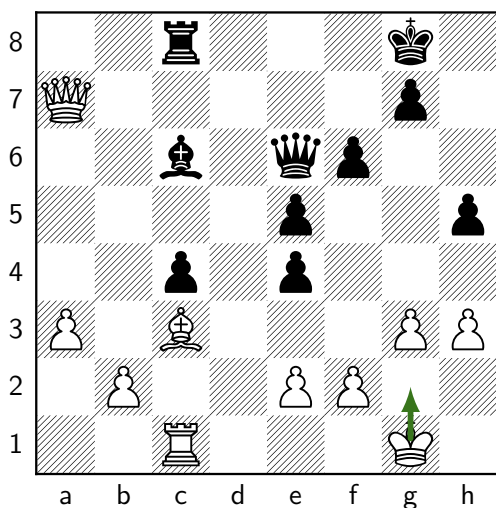
Figure 37: **Move 46, White: b4**. (a) Our position; Melanie plays b4 (b3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2634` (ply 81) [structural distance $d=79$, QAP obj=21]: the master played Bc8; under μ our *Pb3* plays the role of that game’s *Bb2* (a **cross-type** correspondence); cross-type roles: our *Bc2* as their *Qd7*, our *Ra1* as their *Pf7*; transferred move b3b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3066` (ply 75) [structural distance $d=80$, QAP obj=20]: the master played c2; under μ our *Pb3* plays the role of that game’s *Pc6*; cross-type roles: our *Pf4* as their *Re8*, our *bh4* as their *Pg6*; transferred move b3b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3731` (ply 82) [structural distance $d=92$, QAP obj=19]: the master played Bd2; under μ our *Pb3* plays the role of that game’s *Be3* (a **cross-type** correspondence); cross-type roles: our *Ph3* as their *Rb2*, our *Ra1* as their *Pc3*; transferred move b3b4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to b4 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



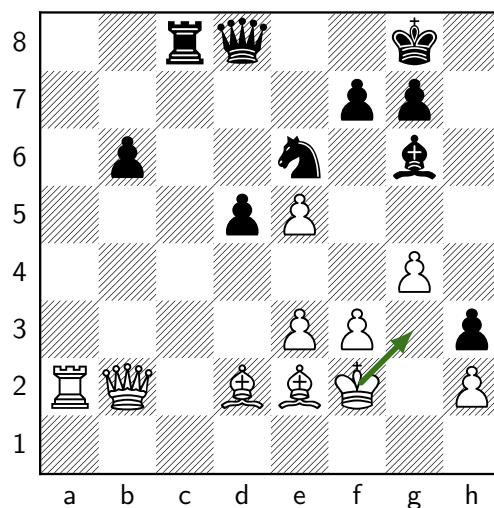
(a)



(b)

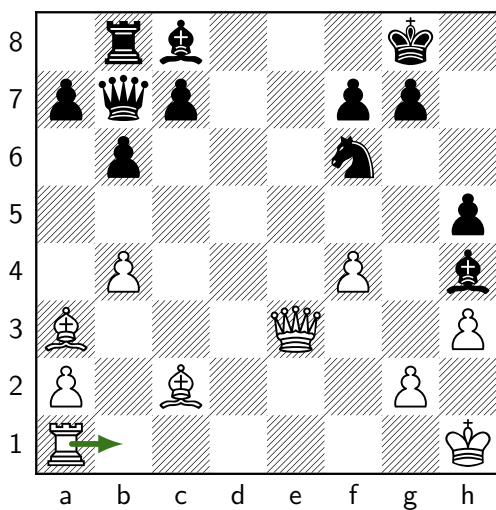


(c)

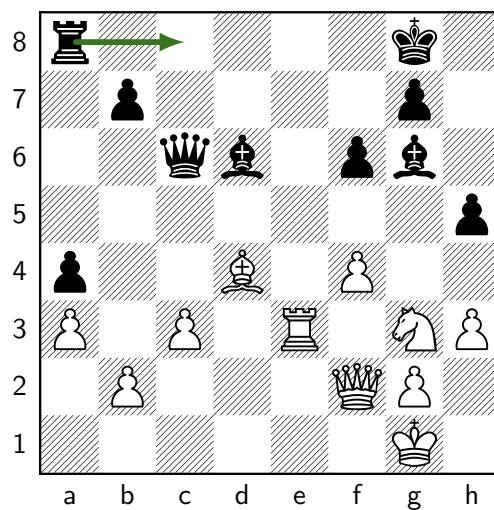


(d)

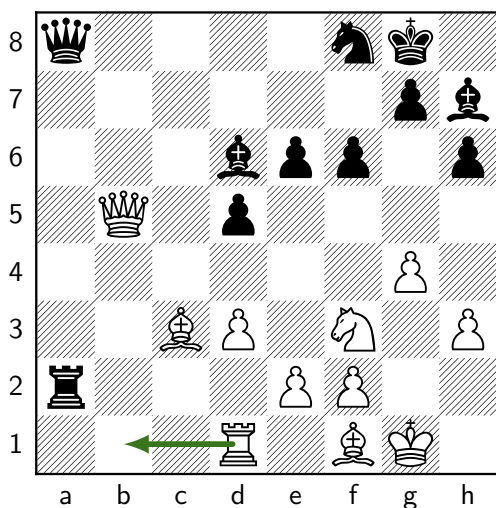
Figure 38: **Move 47, White: Kh1.** (a) Our position; Melanie plays Kh1 (h2h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2308` (ply 88) [structural distance $d=88$, QAP obj=43]: the master played Kh2; under μ our Kh2 plays the role of that game's Kg1; cross-type roles: our bh4 as their Qb2, our pa7 as their Bd2; transferred move h2h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2914` (ply 92) [structural distance $d=89$, QAP obj=33]: the master played Kg2; under μ our Kh2 plays the role of that game's Kg1; cross-type roles: our bh4 as their Pe2, our nf6 as their pe4; transferred move h2h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2209` (ply 78) [structural distance $d=75$, QAP obj=20]: the master played Kg3; under μ our Kh2 plays the role of that game's Kf2; cross-type roles: our Ba3 as their Ra2, our Ph3 as their Bd2; transferred move h2h1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh1 (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



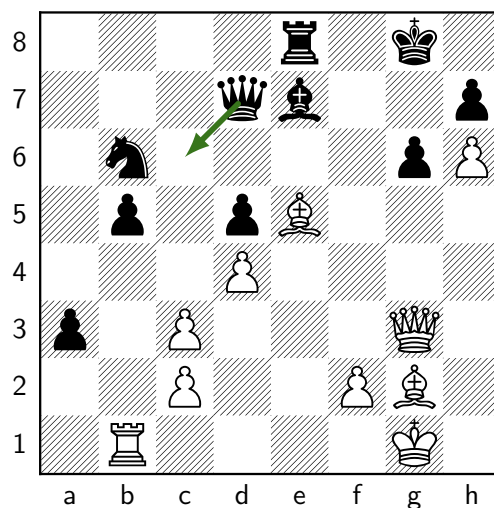
(a)



(b)

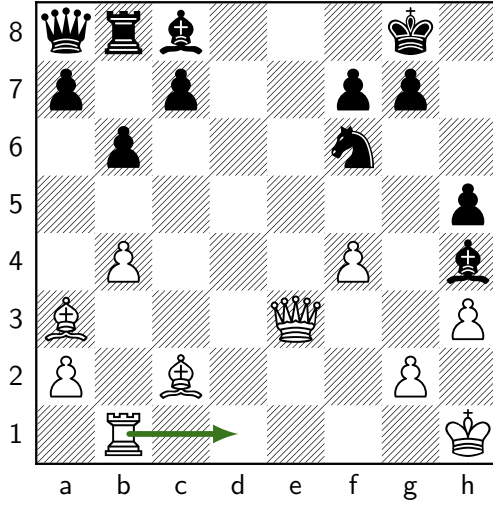


(c)

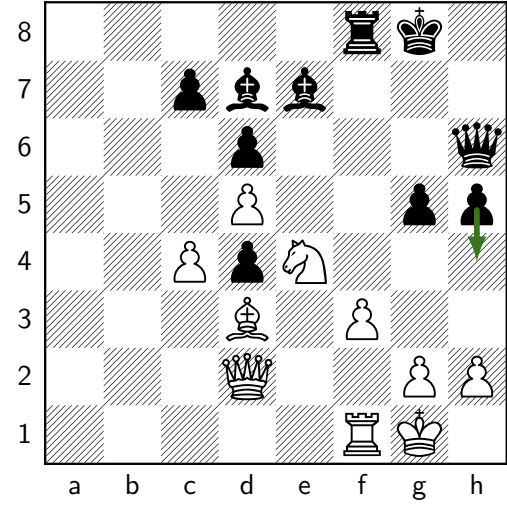


(d)

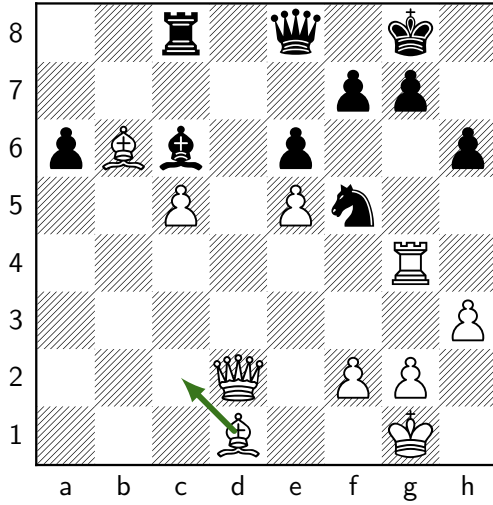
Figure 39: **Move 48, White: Rb1.** (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2967` (ply 93) [structural distance $d=79$, QAP obj=51]: the master played Rc8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *rb8* as their *Pb7*, our *bc8* as their *Qc6*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#919` (ply 78) [structural distance $d=88$, QAP obj=41]: the master played Rb1; under μ our *Ra1* plays the role of that game’s *Rd1*; cross-type roles: our *qb7* as their *ra2*, our *Qe3* as their *Bc3*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1067` (ply 97) [structural distance $d=83$, QAP obj=56]: the master played Qc6; under μ our *Ba3* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *bh4* as their *Nb6*, our *nf6* as their *ph6*; transferred move a3b2. Synthesis: 2 retrieved analogs transferred to Rb1 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



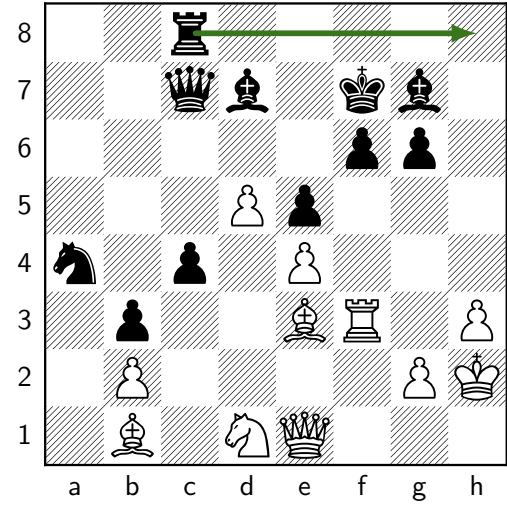
(a)



(b)

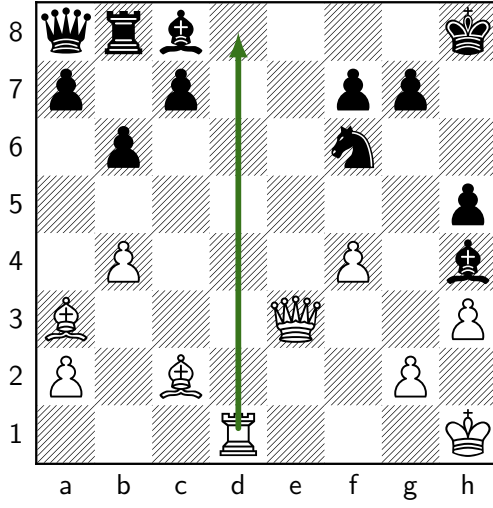


(c)

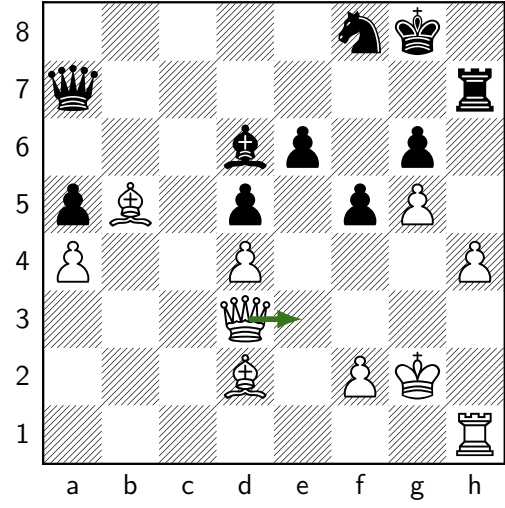


(d)

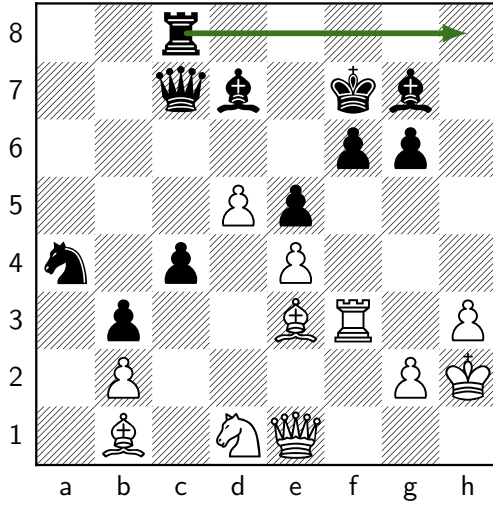
Figure 40: **Move 49, White: Rd1.** (a) Our position; Melanie plays Rd1 (b1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2468` (ply 91) [structural distance $d=80$, QAP obj=52]: the master played h4; under μ our *Rb1* plays the role of that game’s *Ph4* (a **cross-type** correspondence); cross-type roles: our *qa8* as their *Rf8*, our *rb8* as their *Be7*; transferred move b1d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2366` (ply 106) [structural distance $d=73$, QAP obj=50]: the master played Bc2; under μ our *Rb1* plays the role of that game’s *Bd1* (a **cross-type** correspondence); cross-type roles: our *rb8* as their *Ph3*, our *qa8* as their *Rg4*; transferred move b1d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#804` (ply 95) [structural distance $d=78$, QAP obj=41]: the master played Rh8; under μ our *Rb1* plays the role of that game’s *Rc1*; cross-type roles: our *pc7* as their *Qc7*, our *pf7* as their *Bg7*; transferred move b1d1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rd1 (summed QAP objective 171, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



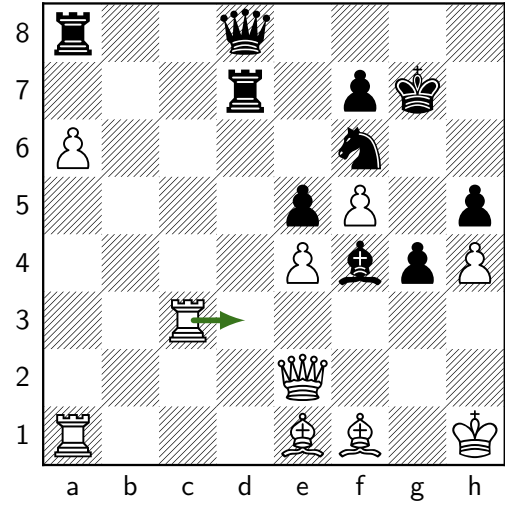
(a)



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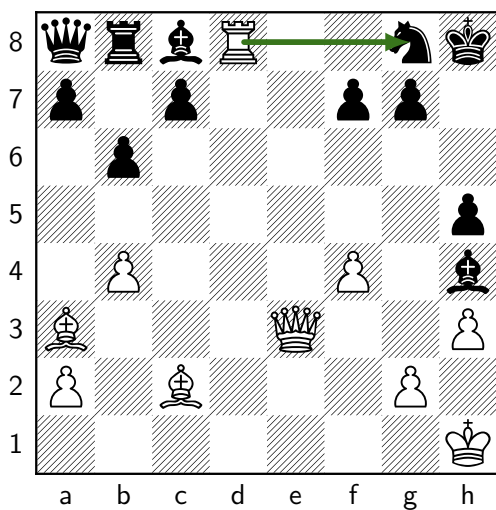


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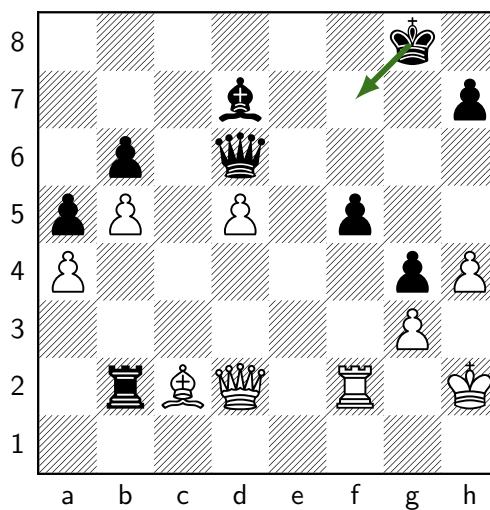


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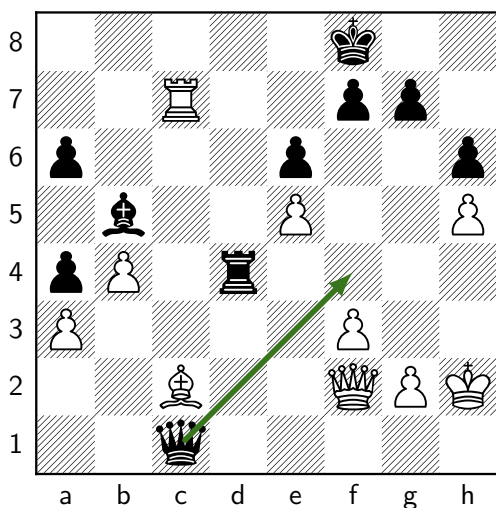
Figure 41: **Move 50, White: Rd8+.** (a) Our position; Melanie plays Rd8+ (d1d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#748` (ply 82) [structural distance $d=86$, QAP obj=45]: the master played Qe3; under μ our Qe3 plays the role of that game’s Qd3; cross-type roles: our Bc2 as their Rh1, our bh4 as their pd5; transferred move e3d3. (c) Retrieved from `ccr1_top_p0003#804` (ply 95) [structural distance $d=83$, QAP obj=43]: the master played Rh8; under μ our Rd1 plays the role of that game’s Rc1; cross-type roles: our Ba3 as their Qc7, our bh4 as their Pg6; transferred move d1b1. (d) Retrieved from `ccr1_top_p0001#39` (ply 84) [structural distance $d=85$, QAP obj=38]: the master played Rd3; under μ our Pb4 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Pf4 as their Bf1, our nf6 as their bf4; transferred move b4b5. Synthesis: no retrieved analog transferred to Rd8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



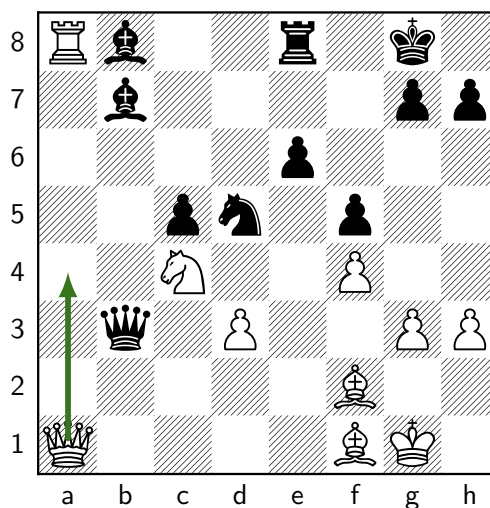
(a)



(b)

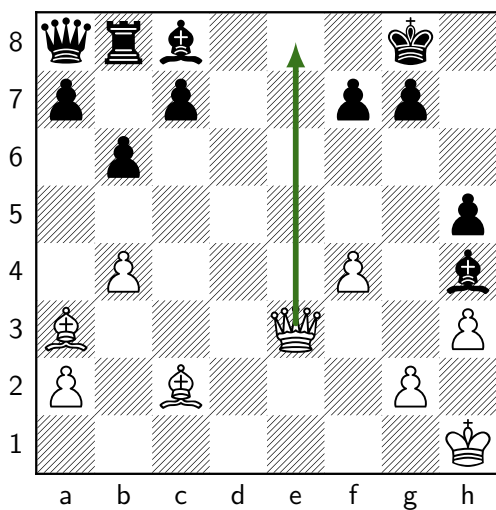


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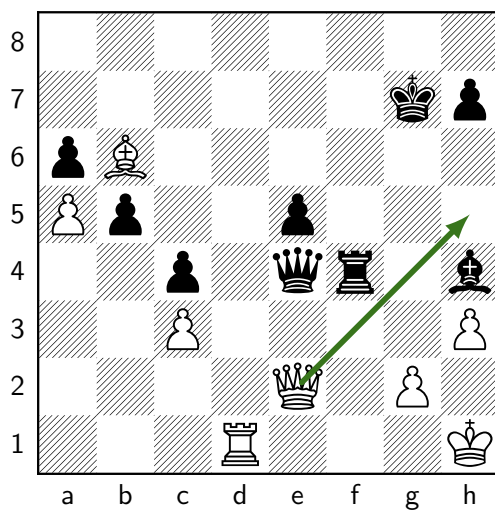


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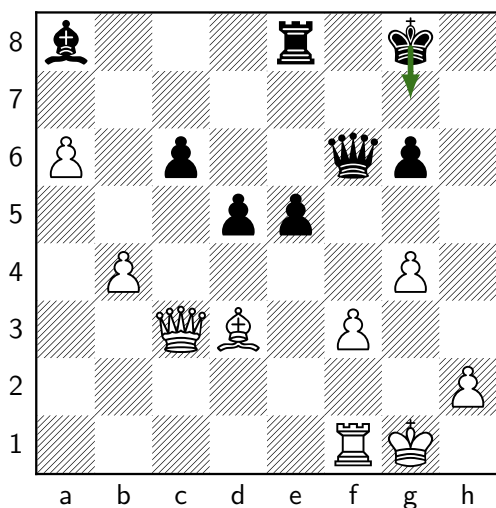
Figure 42: **Move 51, White: Rxd8+.** (a) Our position; Melanie plays Rxd8+ (d8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2435` (ply 103) [structural distance $d=91$, QAP obj=75]: the master played Kf7; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Qe3 as their Bd7, our Bc2 as their Ph7; transferred move h1h2. (c) Retrieved from `ccr1_top_p0001#1970` (ply 89) [structural distance $d=90$, QAP obj=66]: the master played Qf4+; under μ our Rd8 plays the role of that game’s Qc8 (a **cross-type** correspondence); cross-type roles: our Bc2 as their Ph6, our Qe3 as their Bb5; transferred move d8d4. (d) Retrieved from `ccr1_top_p0002#59` (ply 112) [structural distance $d=92$, QAP obj=62]: the master played Qa4; under μ our Qe3 plays the role of that game’s Qa1; cross-type roles: our qa8 as their Bf1, our Rd8 as their qb3; transferred move e3e7. Synthesis: no retrieved analog transferred to Rxd8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



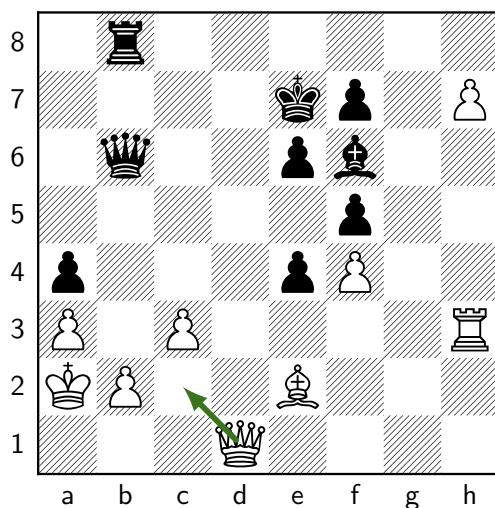
(a)



(b)



(c)



(d)

Figure 43: **Move 52, White: Qe8#.** (a) Our position; Melanie plays Qe8# (e3e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2529` (ply 100) [structural distance $d=69$, QAP obj=15]: the master played Qh5; under μ our Qe3 plays the role of that game's Qe2; cross-type roles: our Ba3 as their Rd1, our Bc2 as their Pa5; transferred move e3e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2550` (ply 89) [structural distance $d=76$, QAP obj=45]: the master played Kg7; under μ our Kh1 plays the role of that game's Kg1; cross-type roles: our Ba3 as their pb4, our Bc2 as their ph2; transferred move h1g1. (d) Retrieved from `ccr1_top_p0000#2880` (ply 88) [structural distance $d=71$, QAP obj=45]: the master played Qc2; under μ our Qe3 plays the role of that game's Qd1; cross-type roles: our Bc2 as their Rh3, our bh4 as their Ph7; transferred move e3d2. Synthesis: 1 retrieved analog transferred to Qe8# (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.